



Department of
Mechanical
Engineering, BUET

ME-400: Project and Thesis

Effects of Laser Powder Bed Fusion Process Parameters on Stress Concentration and Stress Intensity Factor in 17-4PH Stainless Steel

Supervised By:
Dr. Mohammad Motalab
Professor
Dept. of Mechanical Engineering, BUET

Presented By:
Rajib Mostakim (2010062)
Kazi Md Zareer (2010119)

OUTLINE

01

Introduction

Motivation . Objectives .
Scope

02

LPBF Background

Process Parameter .
Materials

03

Background Theory

AM Formulation . SCF . SIF

04

Methodology

ANSYS Chain . Specimens .
Mesh

05

Results & Discussion

SCF & SIF Parametric Study .
Validation

06

Broader Impacts & CEP

Safety . Environment . CEP
mapping

07

Conclusions

Key findings . Novelty .
Future Work

Why This Matters

LPBF is reshaping manufacturing - but residual stress introduces a fracture risk

1

LPBF is transforming manufacturing

- Complex Geometry Build
- Aerospace, Biomedical, Defense, energy sectors.

2

17-4PH: a workhorse alloy

- Precipitation -hardened
- High strength + corrosion resistance.
- Used in valves, shafts, aerospace fittings.

3

The hidden problem

Rapid heating/cooling creates residual stress. This directly affects how parts crack and fail -yet not quantified for parametric process design.

This research quantifies how LPBF process parameters control fracture behavior.

Research Gap

present study combines all three: parametric variation + SCF/SIF extraction + RS → Fracture link

Study	Material	Parametric Variation	SCF / SIF Extraction	RS → Fracture
Becker et al.	Multiple	✗	✗	✗
Leuders et al.	Ti-6Al-4V	✗	✗	✗
Yadollahi et al.	17-4PH	✗	✗	✗
Paul et al.	AlSi10Mg	✓	✗	✗
Parry et al.	Ti-6Al-4V	✓	✗	✗
Present Study	17-4 PH-SS	✓	✓	✓

Objectives

No published work combines all three: parametric variation + SCF/SIF extraction + RS → Fracture link

01

Simulate residual stress fields in LPBF 17-4PH as a function of beam power(P), scan speed (v), hatchspacing (h), layer thickness (t)

02

Extract effective SCF (K_{t}) and Mode -I SIF (K_{I}) from the residual + applied stress field

03

Compare fracture metric sensitivity

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Mesh

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LPBF Process & Four Key Parameters

Laser scans powder → melts → solidifies → layer added

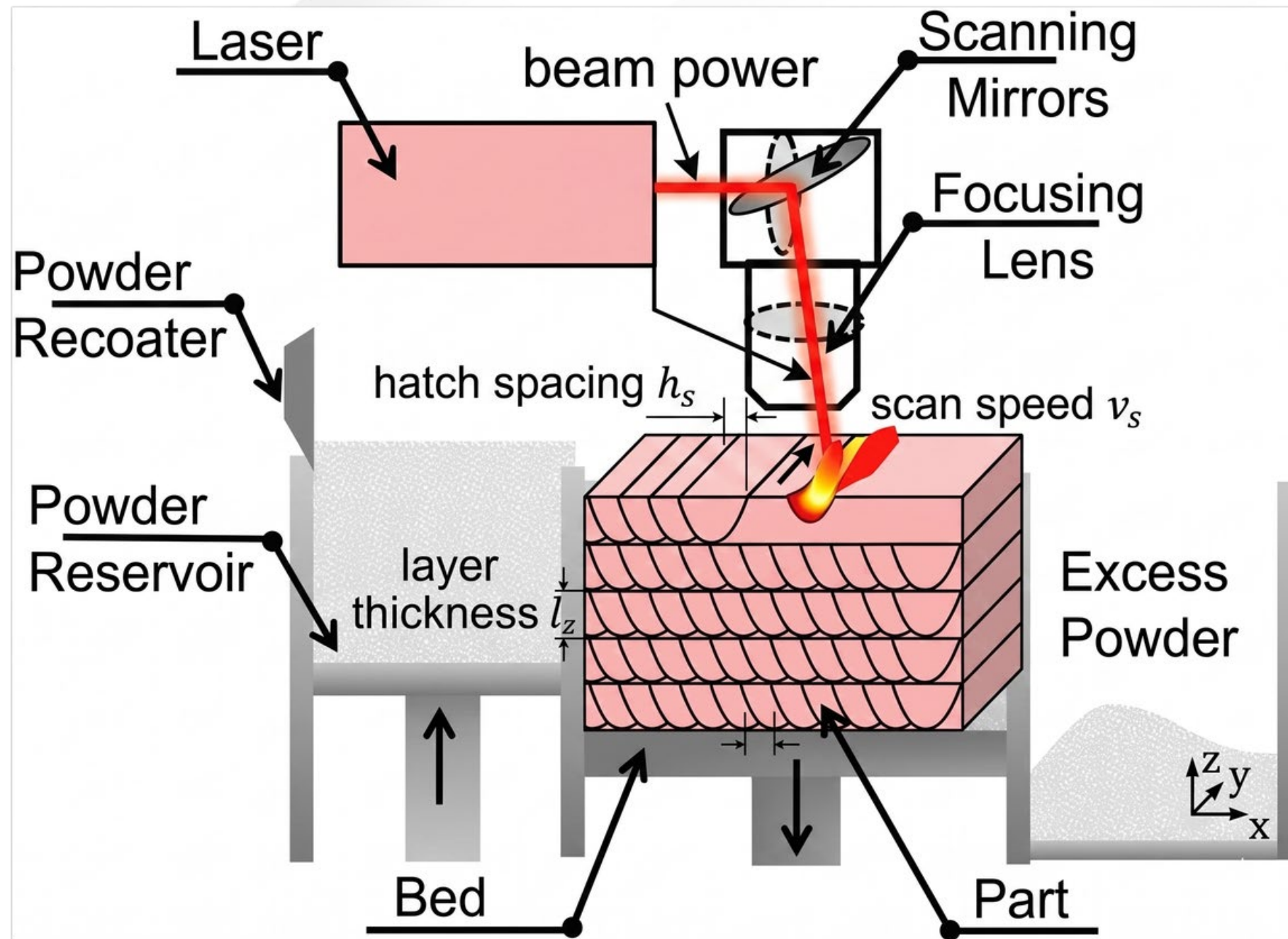


Fig. Laser Bed Powder Fusion (LPBF) Process: Layer
by Layer laser melting

Beam Power (P)

Energy delivered per second

175-325 W

Scan Speed (v)

Velocity of laser beam across powder bed

600- 1800 mm/s

Layer Thickness (t)

Distance between adjacent scan lines

0.08–0.17 mm

Hatch Spacing (h)

Thickness of each deposited powder layer

0.03–0.06 mm

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Theoretical Background - LPBF

What the solver actually solves we use ANSYS Workbench as the FEM solver.

Transient Thermal

Heat conduction equation:

$$\rho c_p \left(\frac{\partial T}{\partial t} \right) = \nabla \cdot (k \nabla T) + Q$$

- ρ, c_p, k : temperature -dependent material properties
- Q : deposited heat, added layer by layer
- Energy enters via layer activation (element birth -and -death), not a tracked laser

Output: temperature history $T(x, y, z, t)$

Static Structural

Mechanical equilibrium:

$$\nabla \cdot \sigma + b = 0$$

Thermo -elastoplastic strain:

$$\epsilon_{\text{total}} = \epsilon_{\text{elastic}} + \epsilon_{\text{plastic}} + \alpha \Delta T$$

- $\alpha \Delta T$ = thermal strain its constraint drives residual stress
- von Mises yield with bilinear isotropic hardening

Output: residual stress field $\sigma_{\text{res}}(x, y, z)$

Both stages are solved by FEM in ANSYS Workbench 2025 R1

Theoretical Background - Fracture Mechanics

Stress concentration and stress intensity extracted on the residual-stressed body.

STRESS CONCENTRATION

$$K_{t, eff} = \frac{\sigma_{max}}{\sigma_{nom}} , \quad \sigma_{nom} = \frac{F}{A_{net}}$$

peak (von Mises) stress taken from the coupled residual + applied field $\rightarrow$ residual exceeding the geometric limit $K_t \leq 3$

-dominated,

STRESS INTENSITY

$$J = \int_{\Gamma} \left(W \delta_{1j} - \sigma_{ij} \frac{\partial u_i}{\partial x_1} \right) n_j d\Gamma$$

$$K_I = \sqrt{E' J} , \quad E' = E / (1 - \nu^2)$$

$$K_{I, total} = K_{I, applied} + K_{I, residual}$$

Mode I · contour / interaction (domain) integral · LEFM superposition of applied and residual parts

SIF from the interaction (domain) integral over crack

-front contours · Mode I

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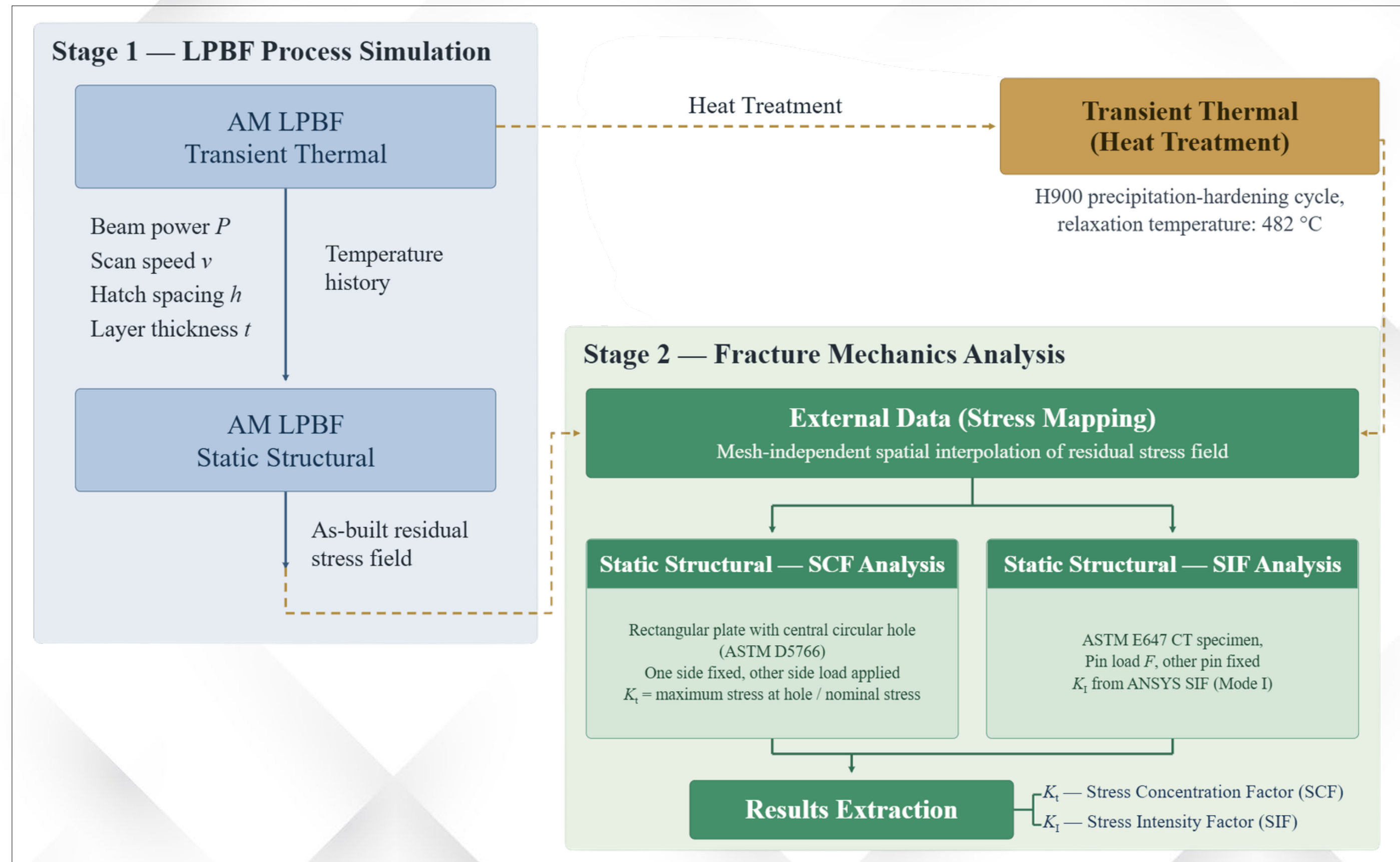
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Simulation Workflow

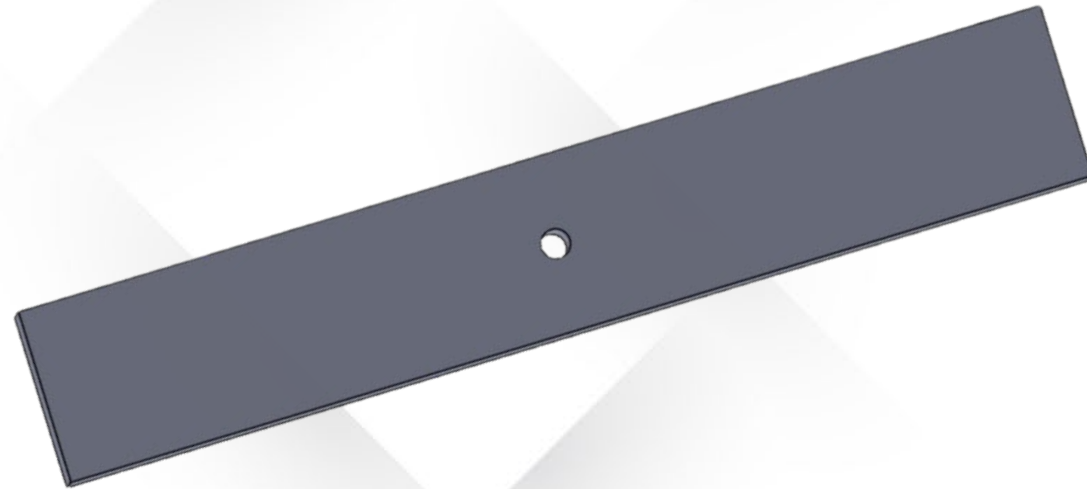
Two-staged coupled simulation (Used Ansys 2025 R1 as a solver)



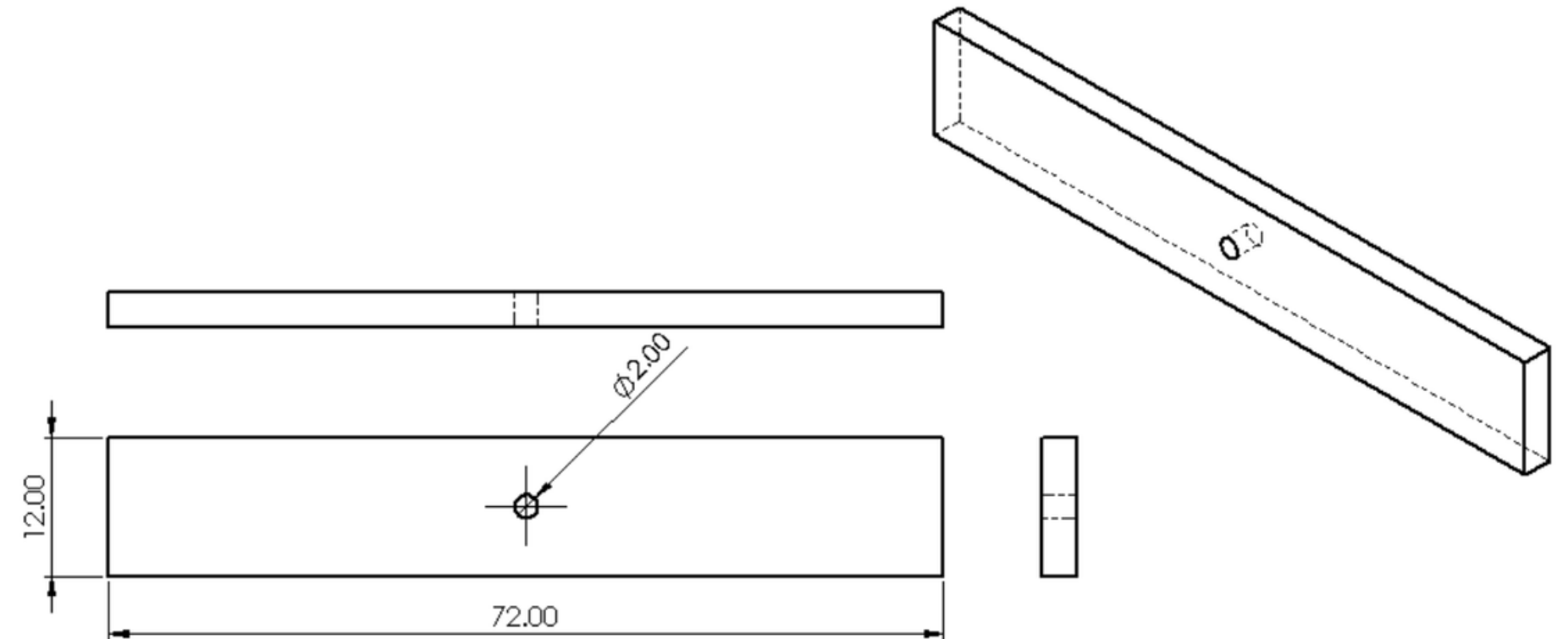
This process is repeated for every parameter combination (OFAT design)

Specimens

SCF Specimen — ASTM D5766

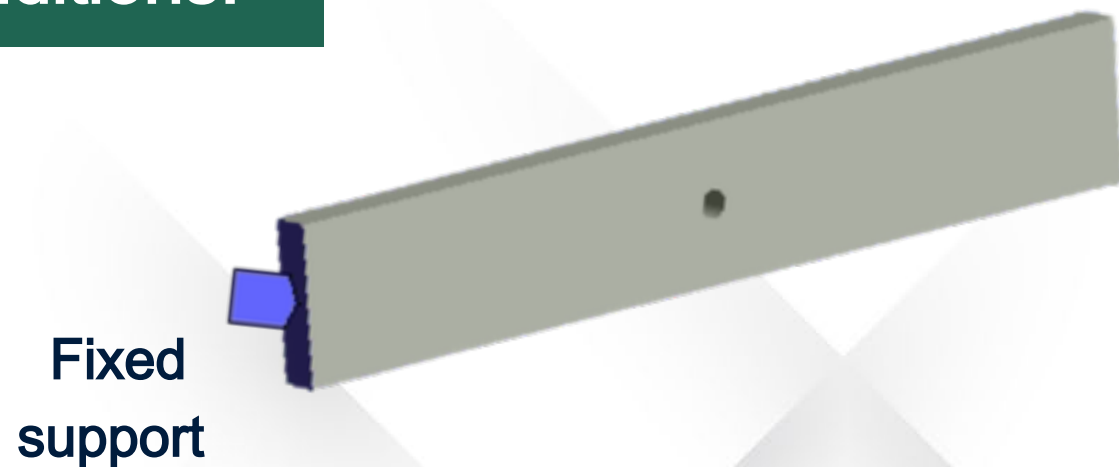


SolidWorks Model

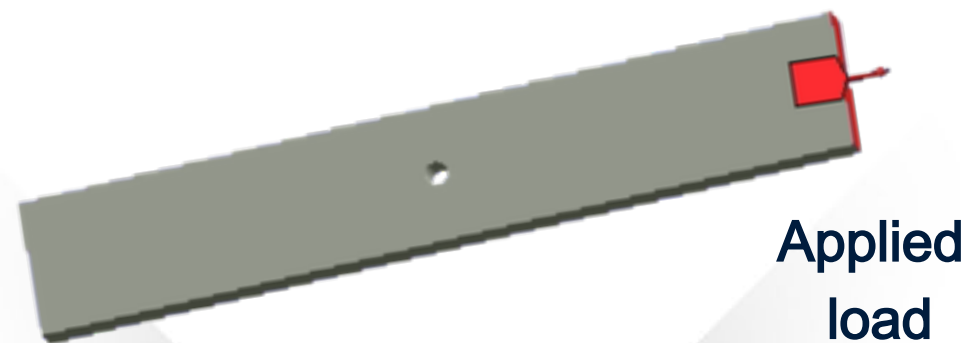


Dimensions: 72 × 12 × 3 mm, 2 mm central hole, $w/D = 6$

Boundary Conditions:



Fixed support



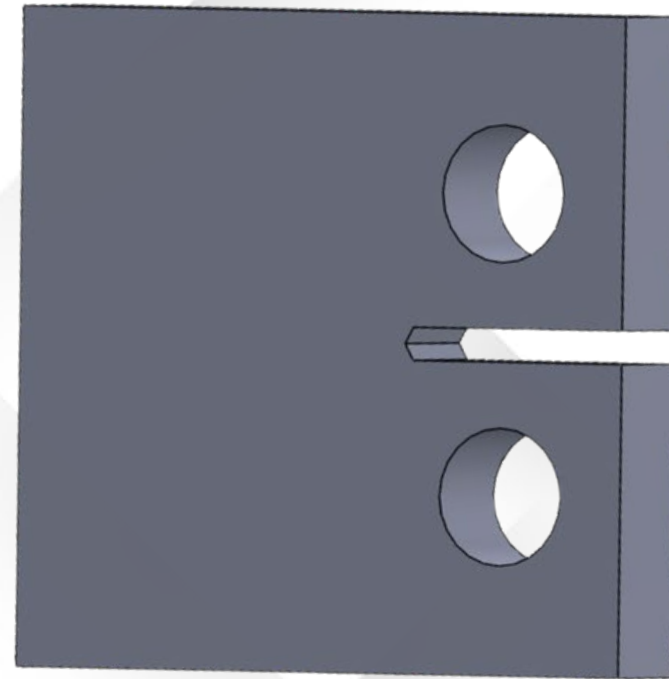
Applied load



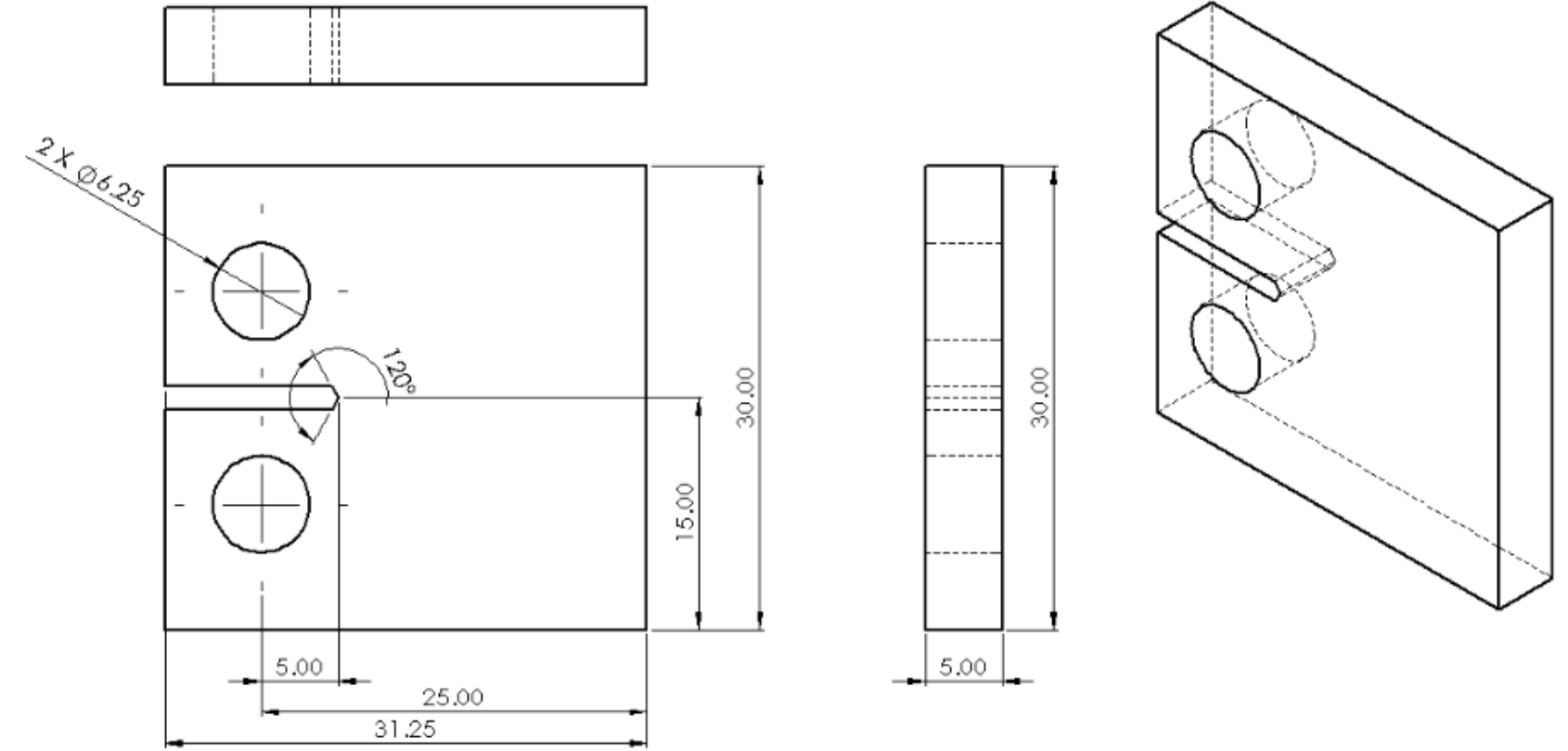
Finite Element mesh

Specimens

SIF Specimen — ASTM E647

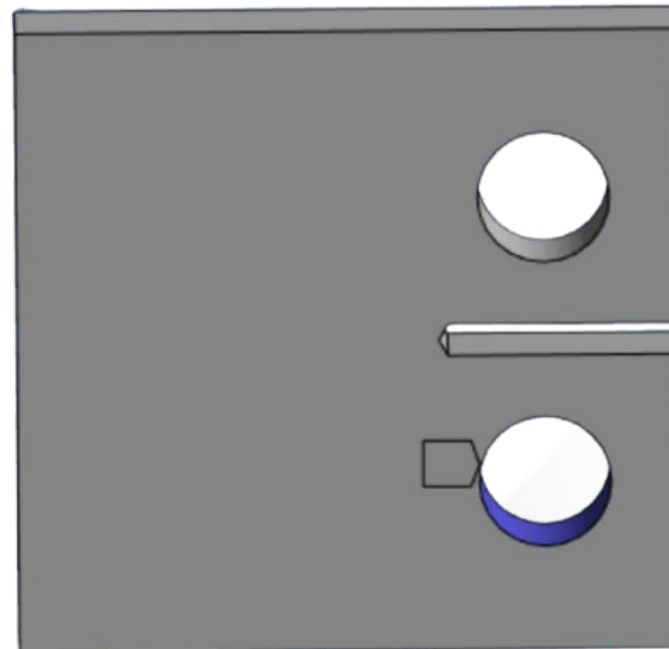


SolidWorks Model

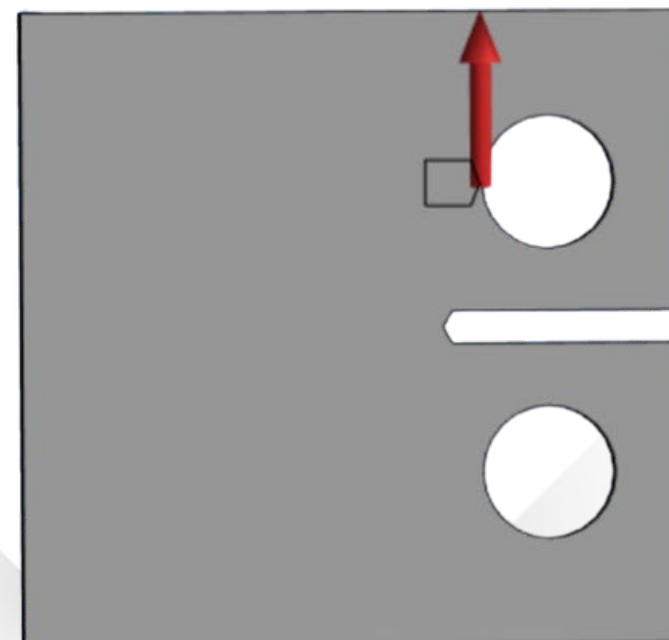


Dimensions: $W=25$ mm, Thickness $B=5$ mm, $a/W=0.2$

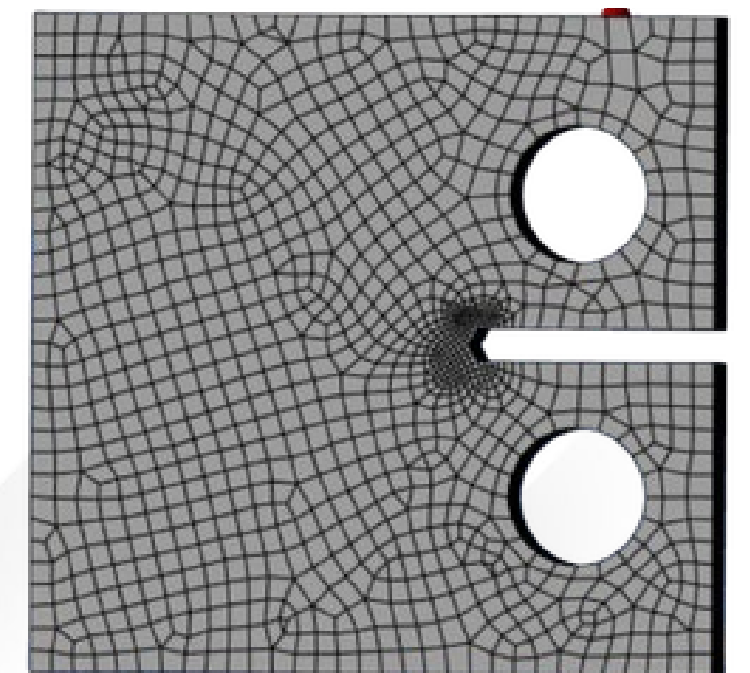
Boundary Conditions:



Fixed Support



Applied Load



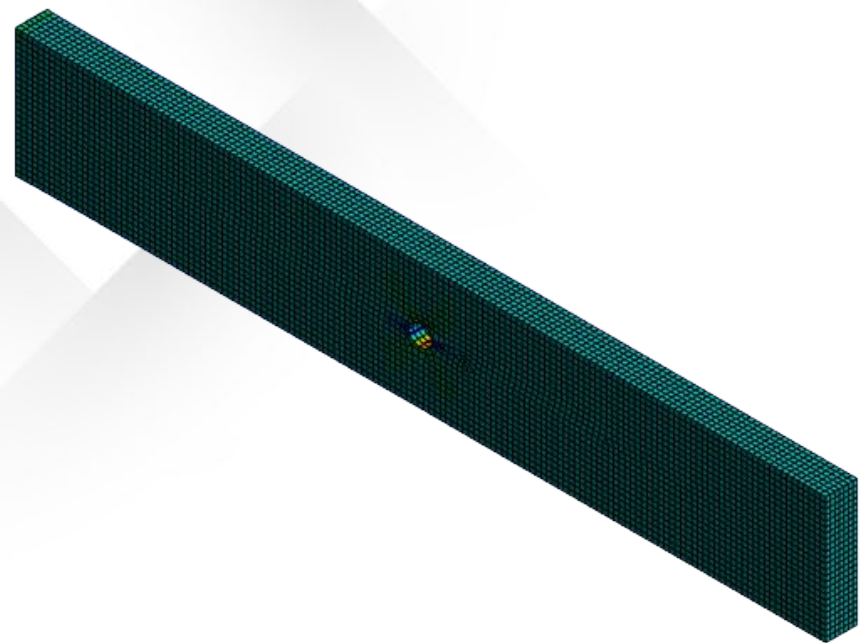
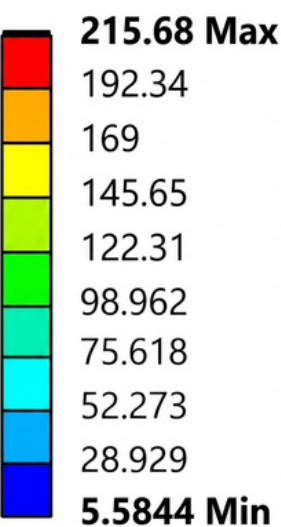
Meshing

SCF and SIF Validation

Validation is done against analytical solution

Stress Concentration Factor (SCF)

Open-Hole Plate · Peterson finite-width solution · σ in MPa



$$K_t = 3 - 3.13(d/W) + 3.66(d/W)^2 - 1.53(d/W)^3 \rightarrow K_t = 2.573$$

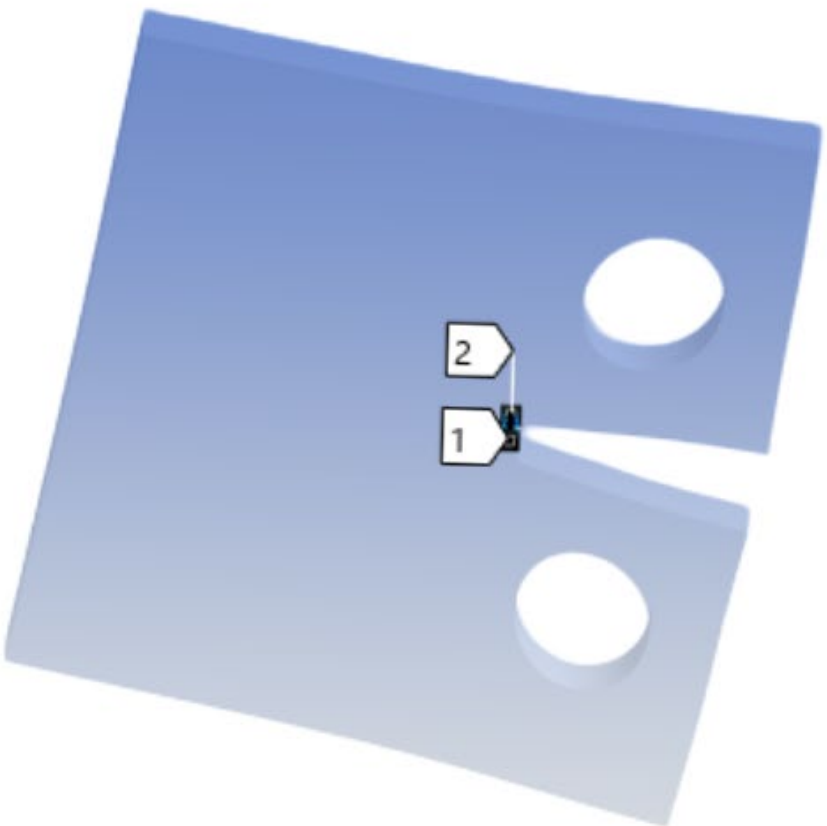
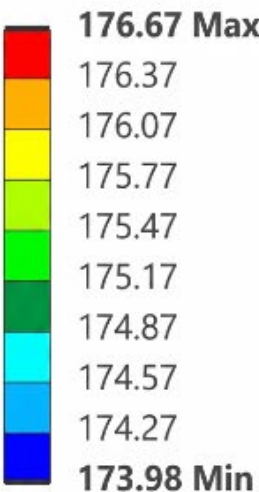
214.4
ANALYTICAL

215.68
FEA (ANSYS)

0.60%
DEVIATION

Stress Intensity Factor (SIF)

Compact Tension · Srawley wide-range solution · KI in MPa·√mm



$$K_I = (P / B\sqrt{W}) \cdot f(a/W), \quad f(0.2) = 4.274, \quad a/W = 0.2$$

170.9
ANALYTICAL

176.67
FEA (ANSYS)

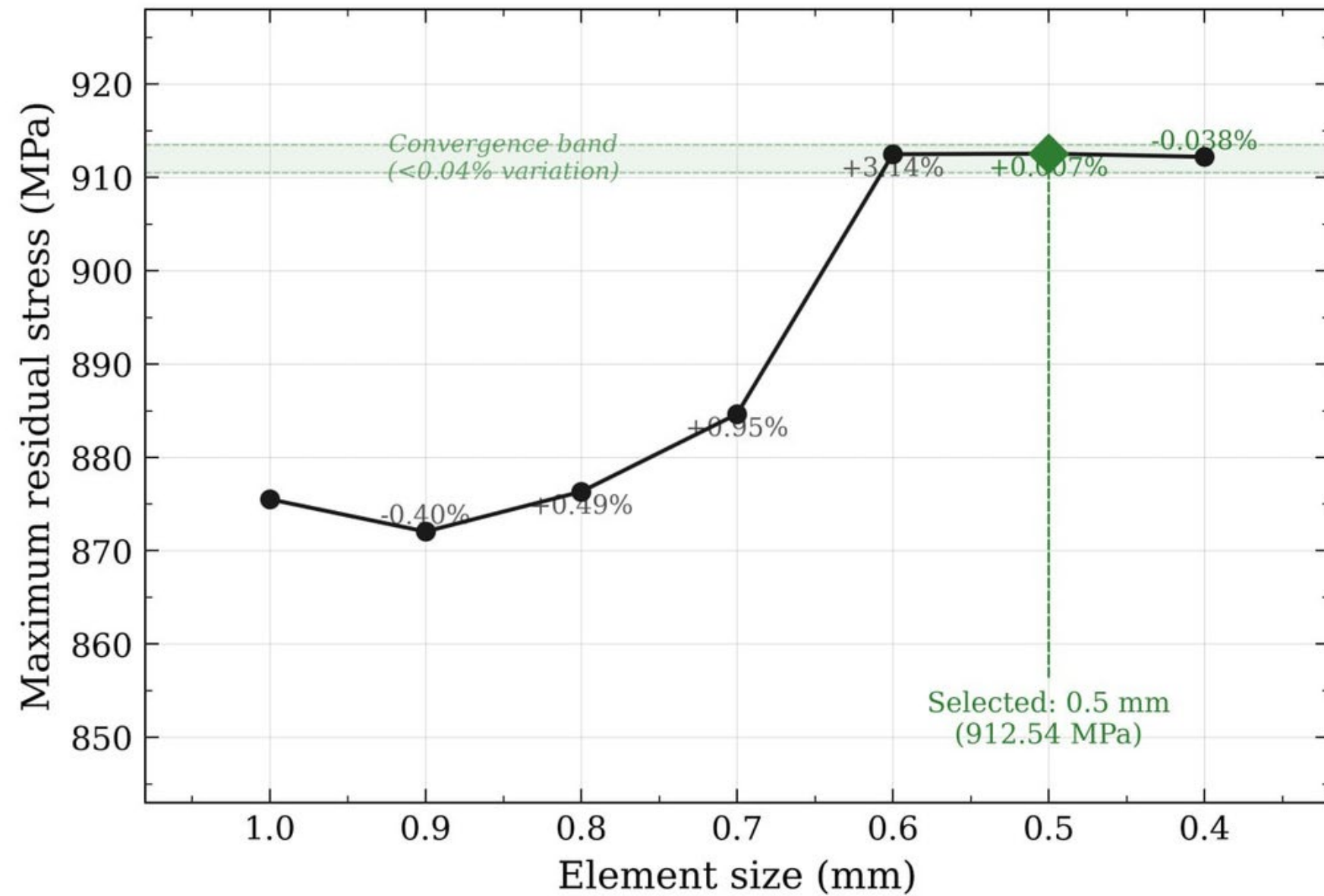
3.38%
DEVIATION

Both deviations fall within the 5% engineering FEA tolerance

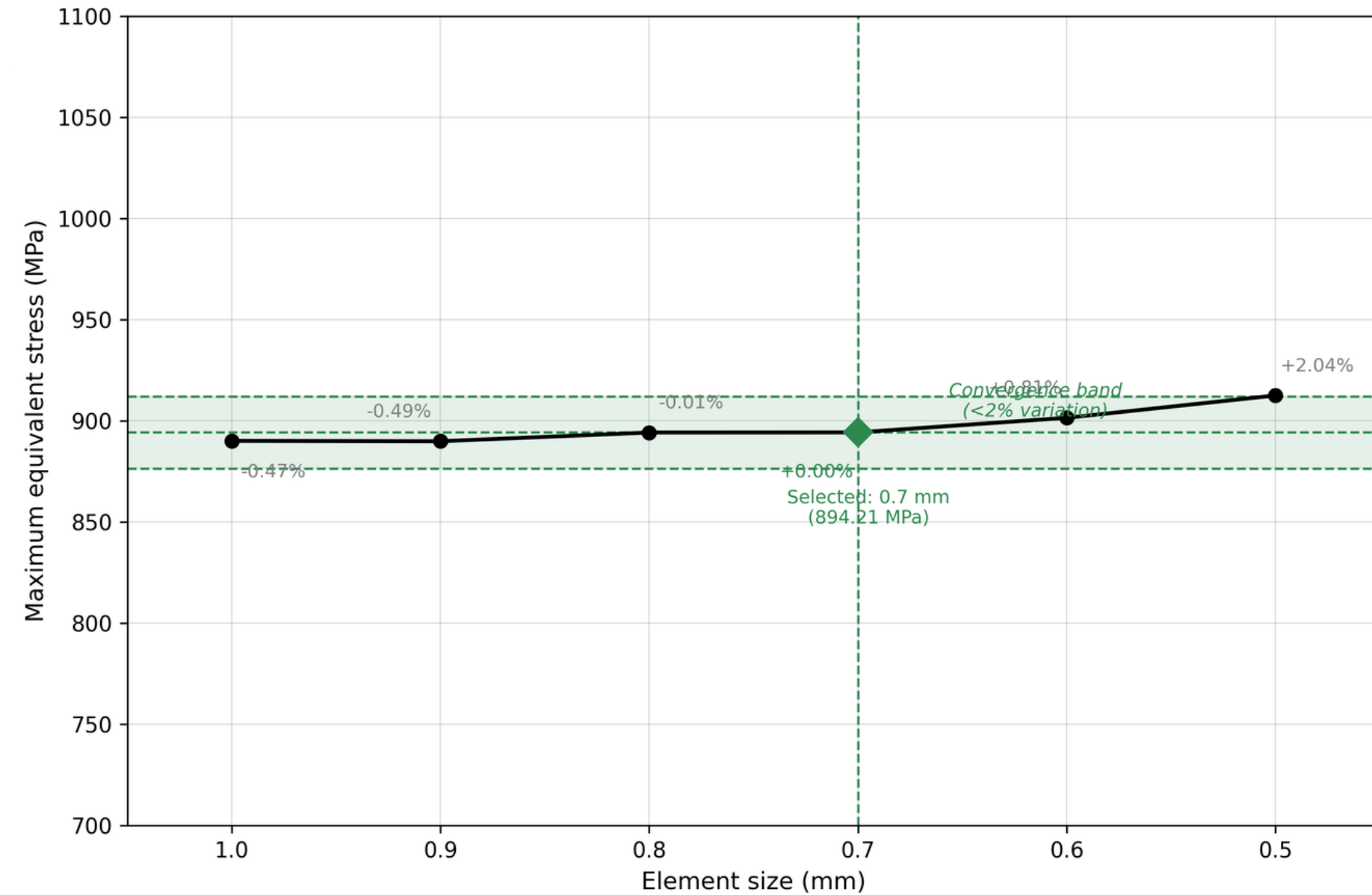
— establishing a verified baseline for the coupled residual

-stress parametric study.

Convergence Study



Residual stress converges at 0.5 mm (plateau: 0.4 – 0.6 mm, <2% variation).
Adopted for all SCF simulations.



Element size converge was taken for 2%. So we take 0.7 mm element size for SIF.

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Mesh

05

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06

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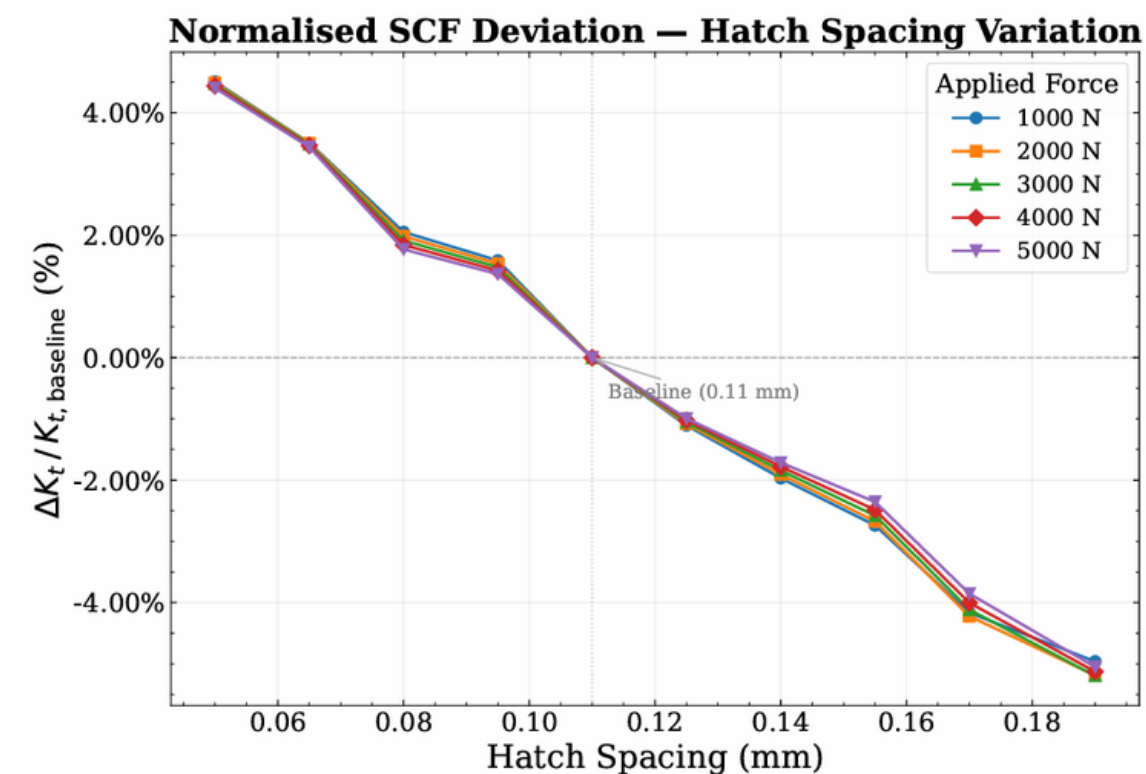
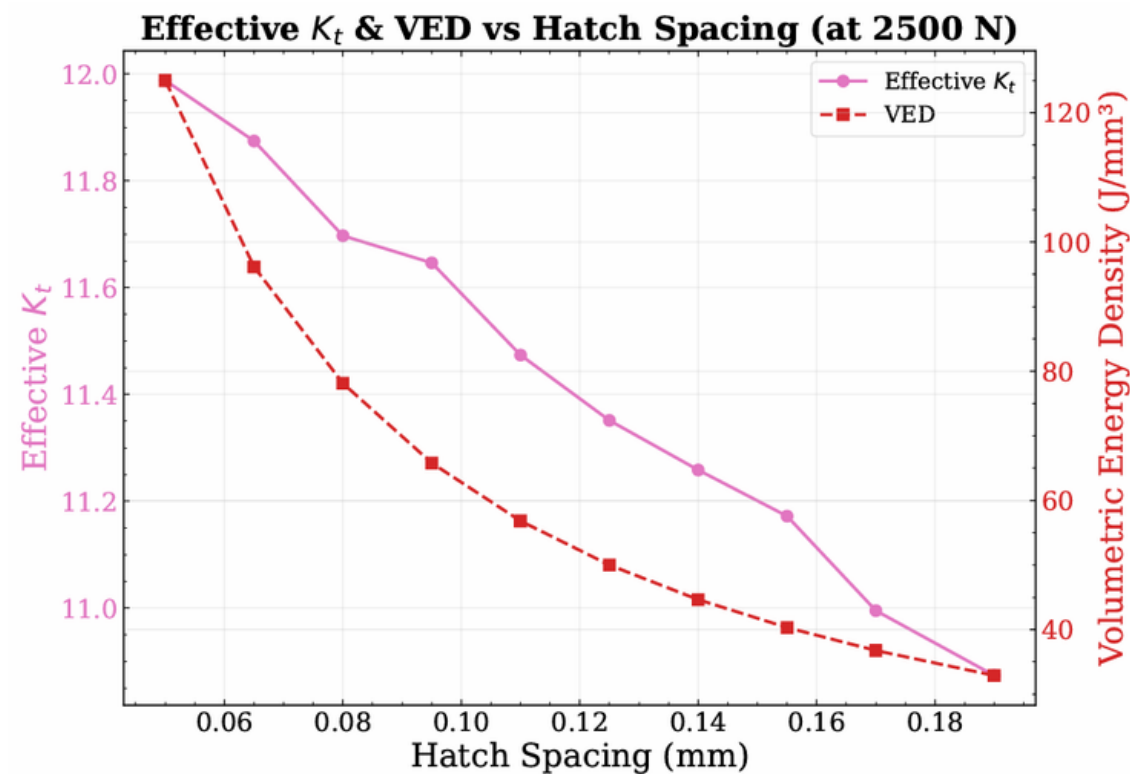
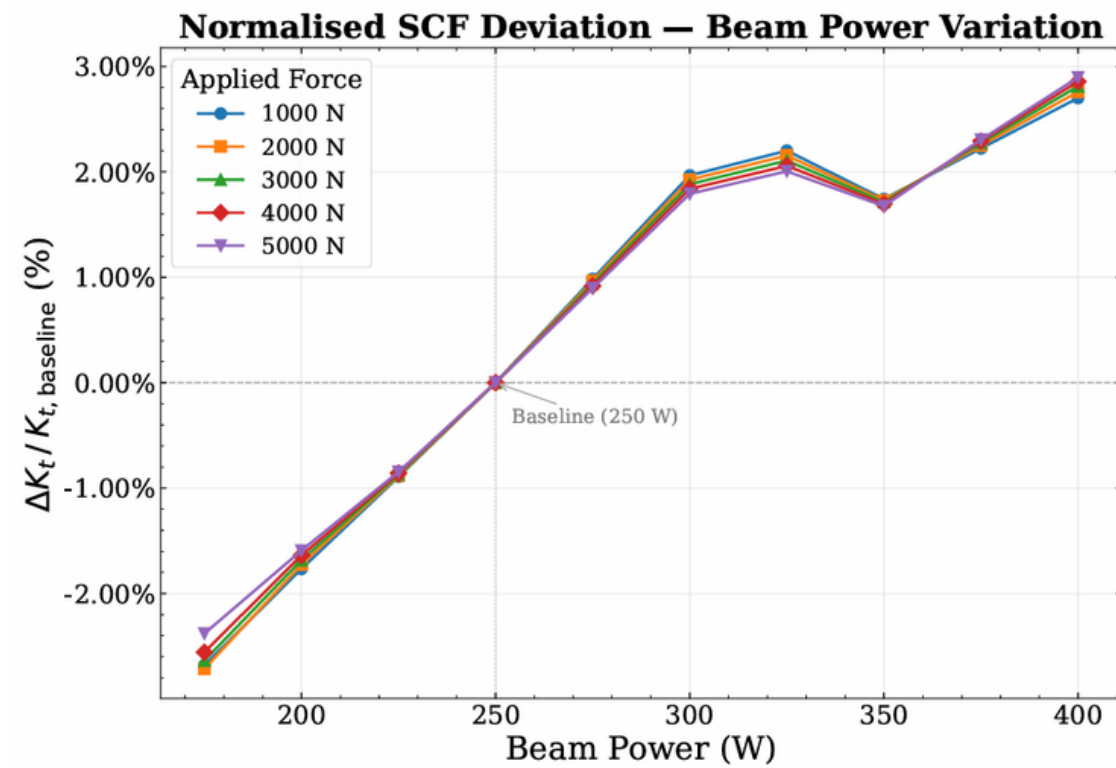
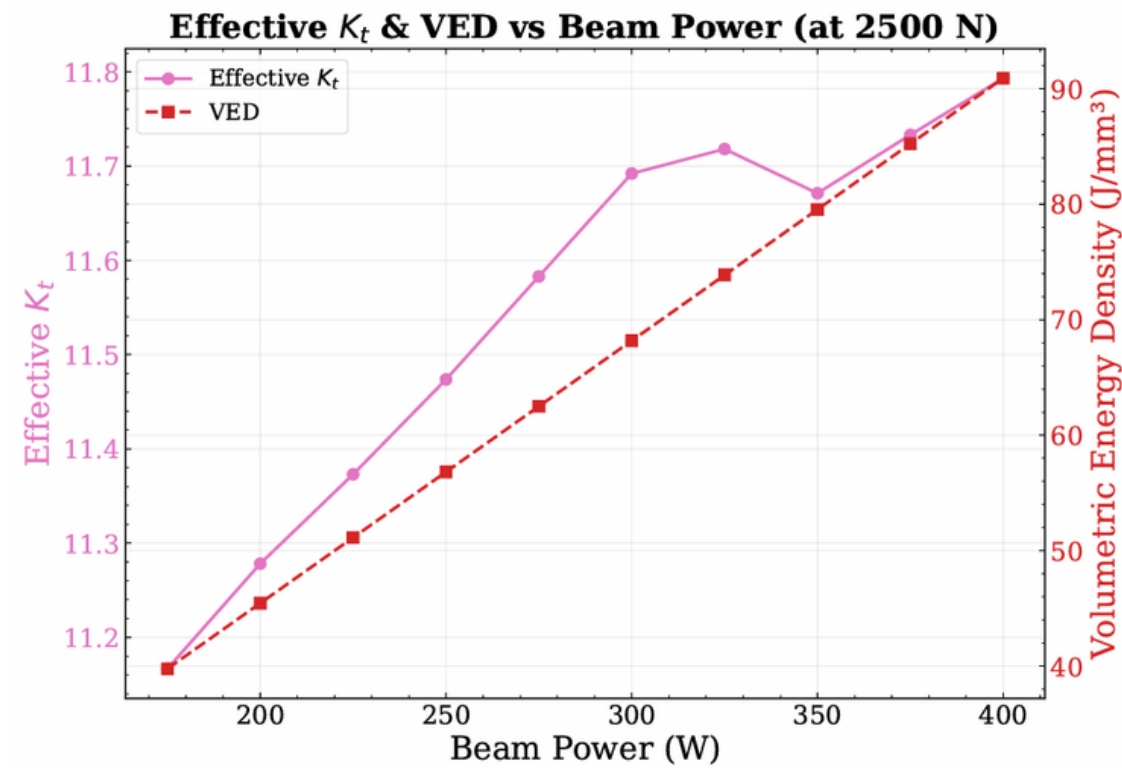
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07

Conclusions

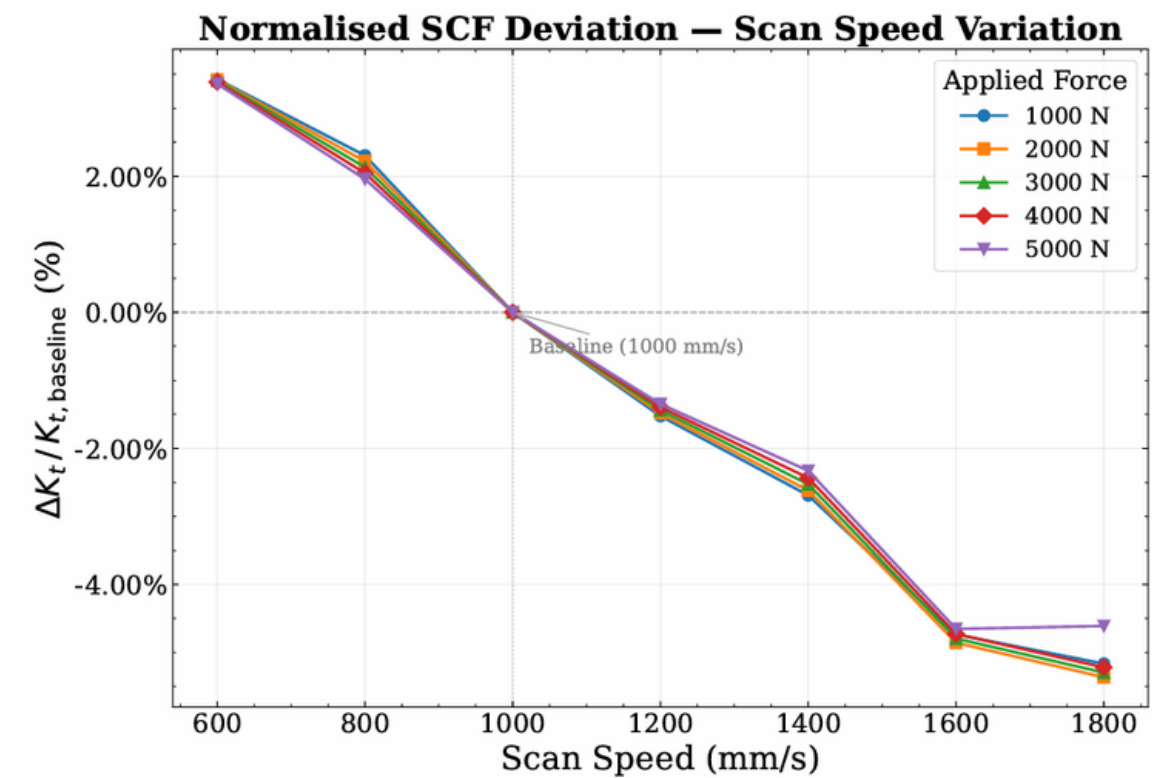
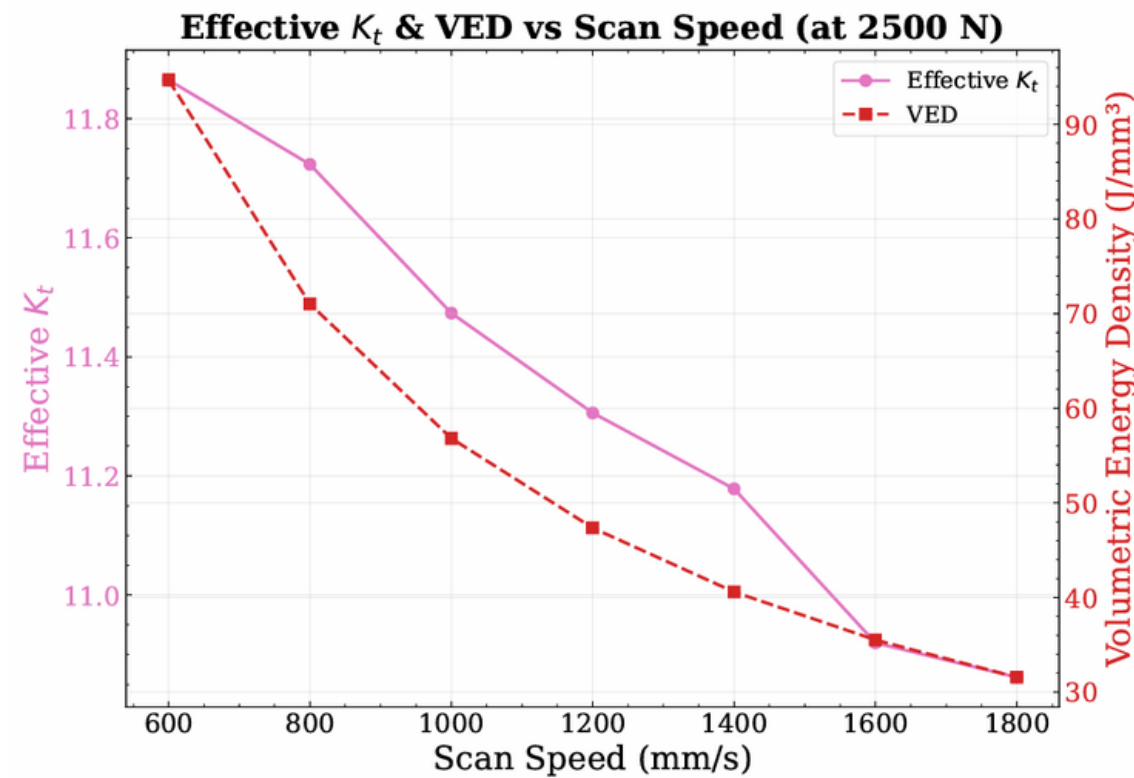
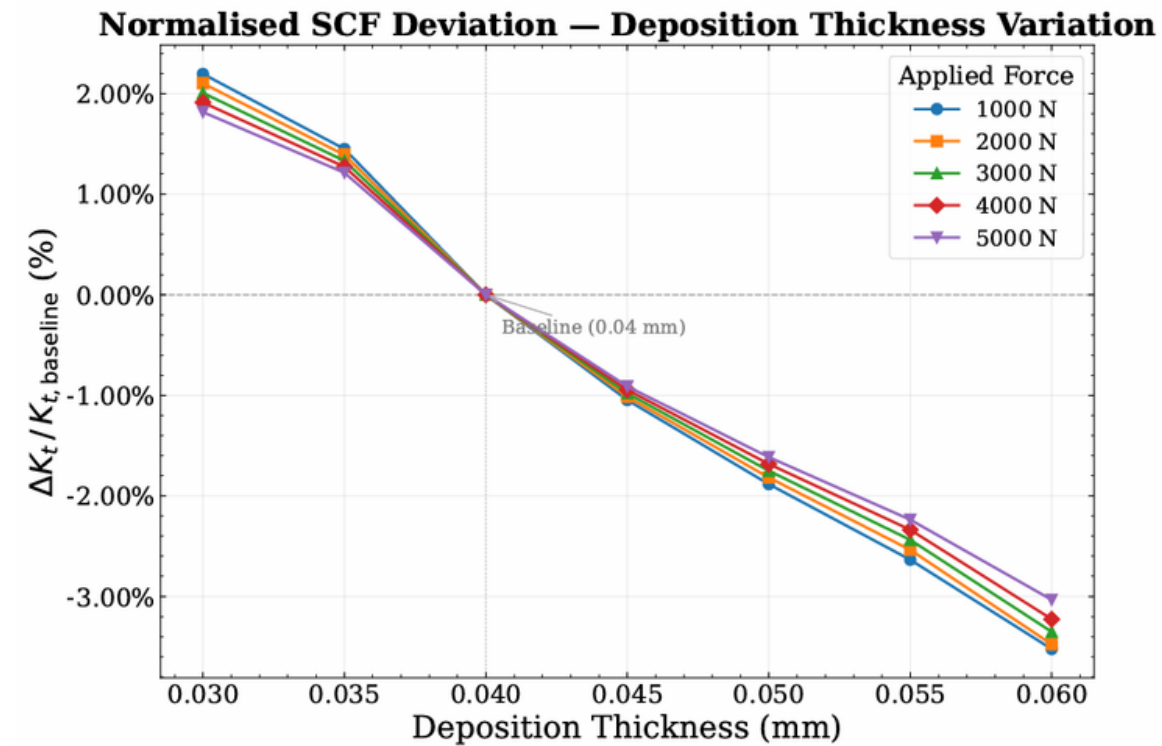
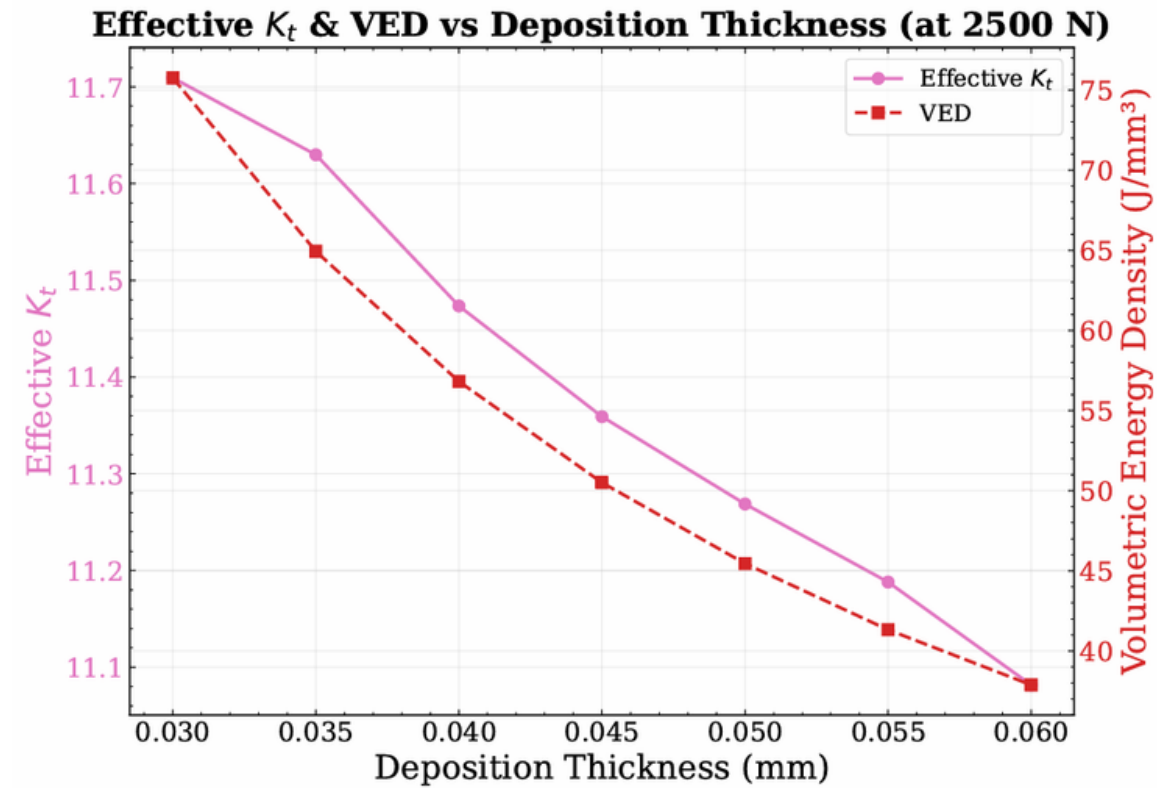
Key findings . Novelty .
Future Work

SCF Results



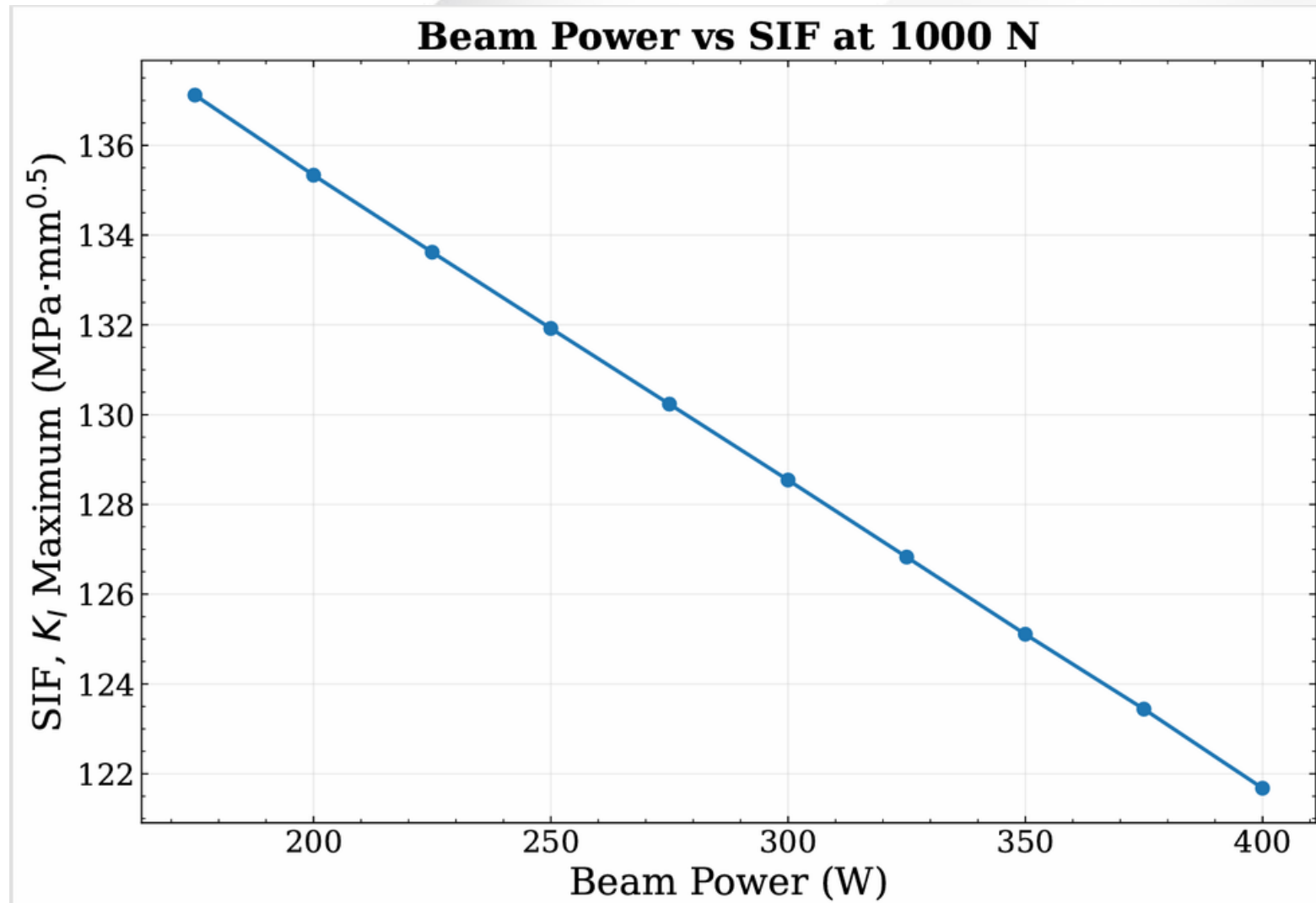
- Beam power $\uparrow \rightarrow K_{t,eff} \uparrow$ (≈ 11.2 to 11.8). More power raises $E_v \rightarrow$ more residual stress.
- Hatch spacing $\rightarrow K_{t,eff} \rightarrow$ (≈ 12.0 to 10.9). Wider hatch lowers $E_v \rightarrow$ less residual stress.
- Beam Power has 5.6% variation
- Hatch spacing has 9.71%
- Residual Stress Dominates Throughout All Force Levels
- Local Dip: On the Transition of Conduction Mode to Key-Hole Mode

SCF Results



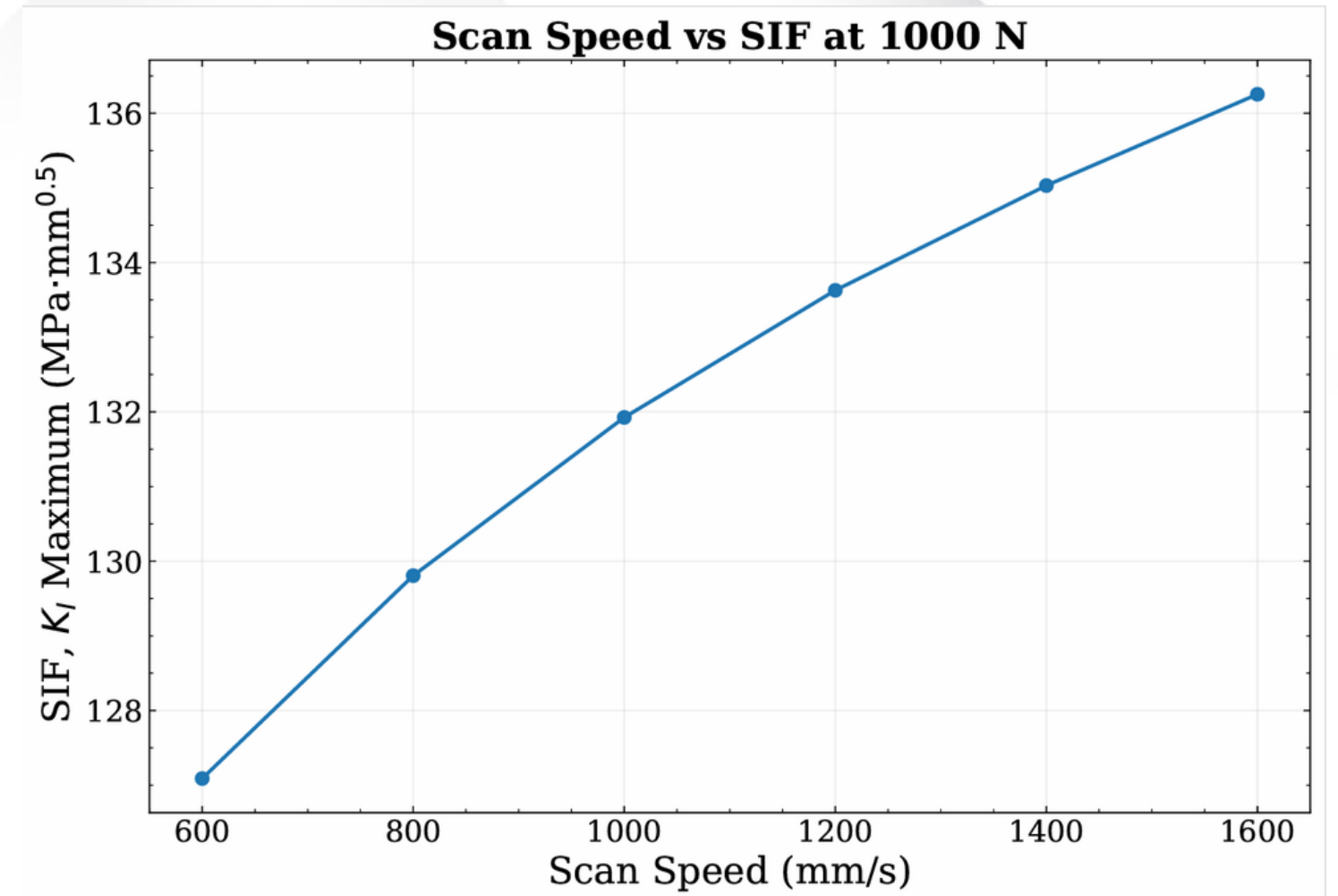
- Layer (deposition) thickness $\uparrow \rightarrow K_{t,eff} \rightarrow (\approx 11.7 \rightarrow 11.1)$. Thicker layers lower E_v .
- Scan speed $\rightarrow \rightarrow K_{t,eff} \rightarrow (\approx 11.85 \rightarrow 10.9)$. Faster scan lowers E_v .
- Deposition Thickness has 5.47%
- Hatch spacing has 8.74%
- Residual Stress Dominates Throughout All Force Levels

SIF Results



SIF Decreases With Beam Power : Crack Tip
Residual Stress Mechanism

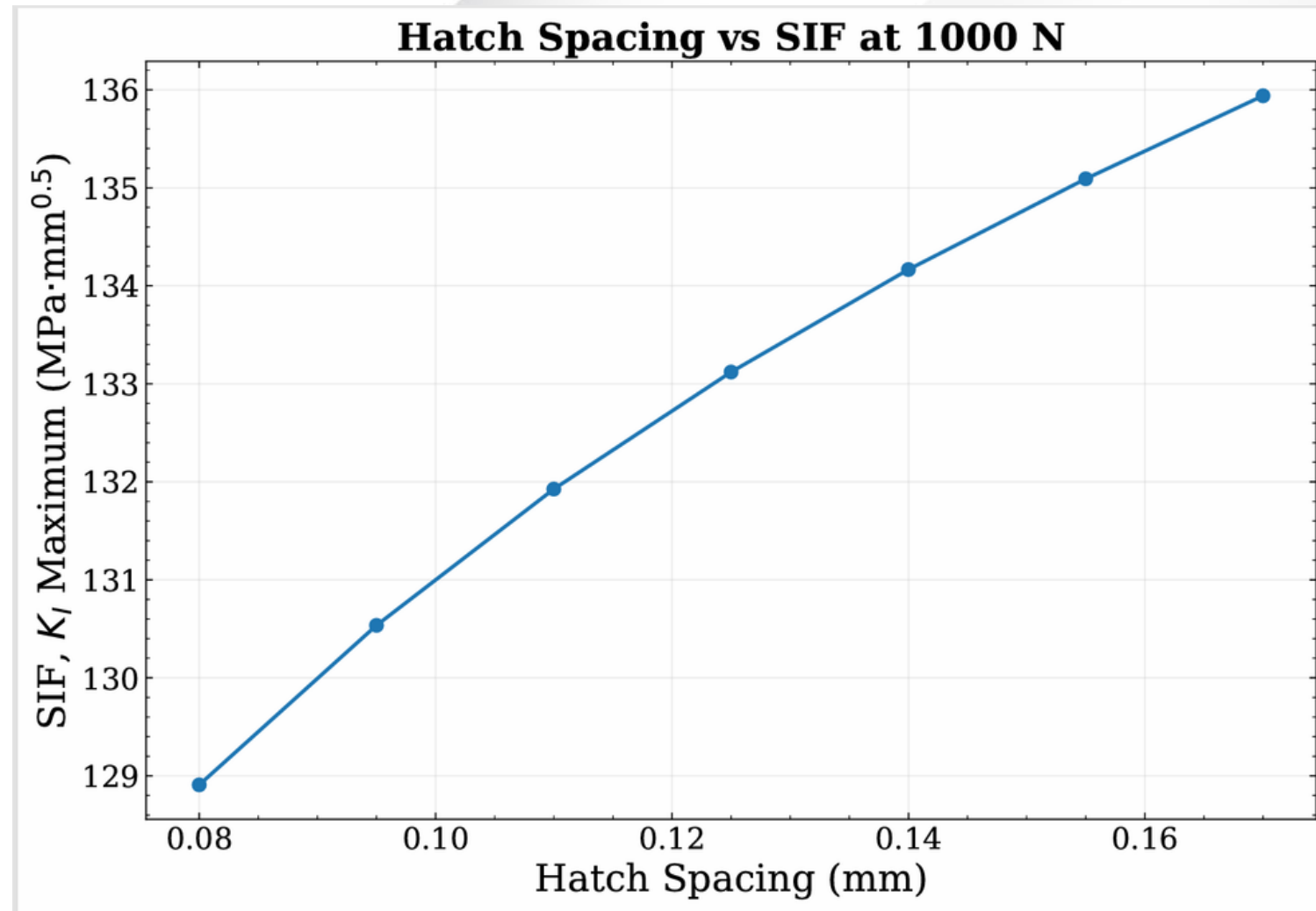
~11.71% variation. Most Dominant



SIF Increases With Scan Speed : Crack Tip
Residual Stress Mechanism

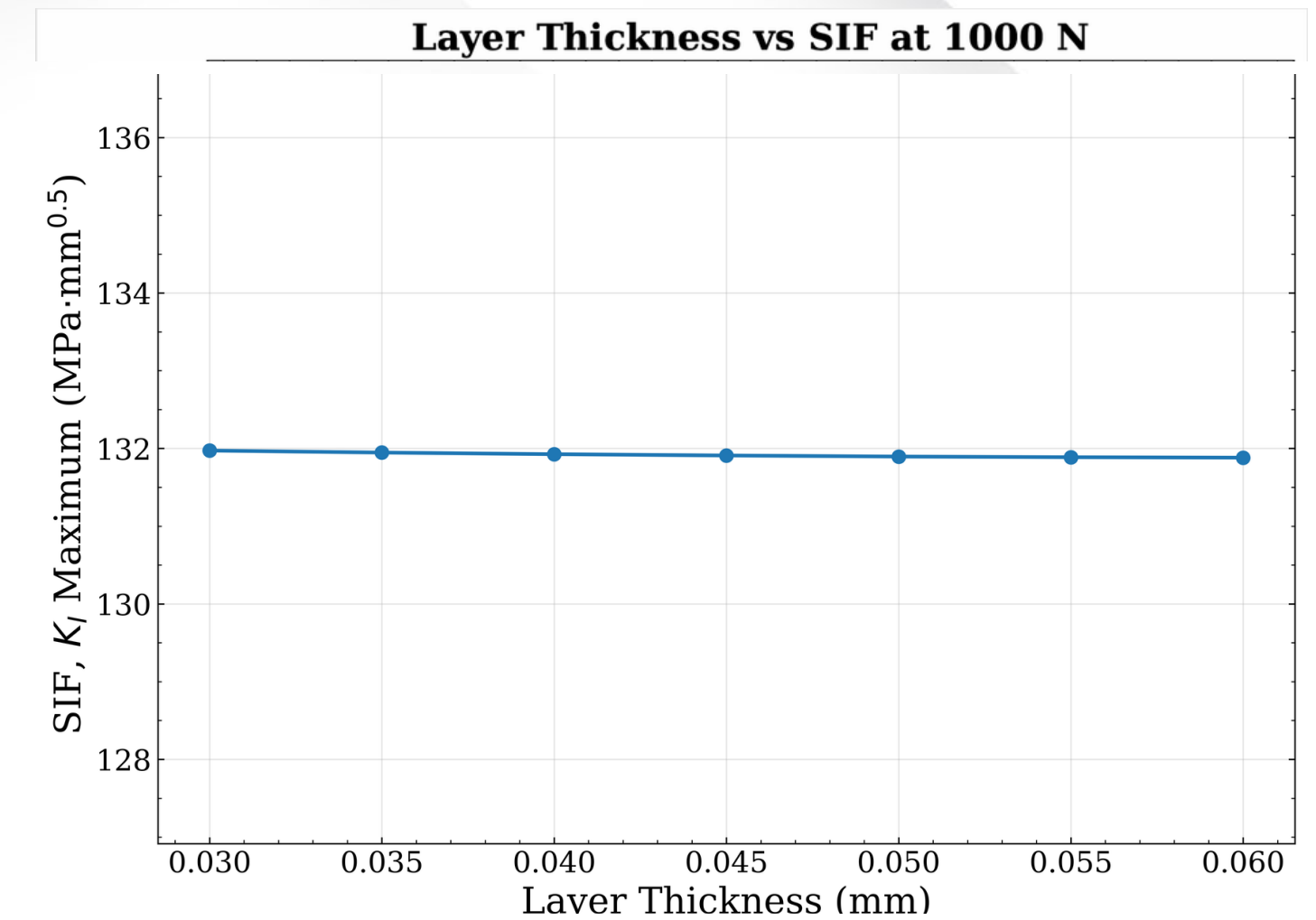
~6.95% variation. Second Most Dominant

SIF Results



SIF Increases With Hatch Spacing : Crack
Tip Residual Stress Mechanism

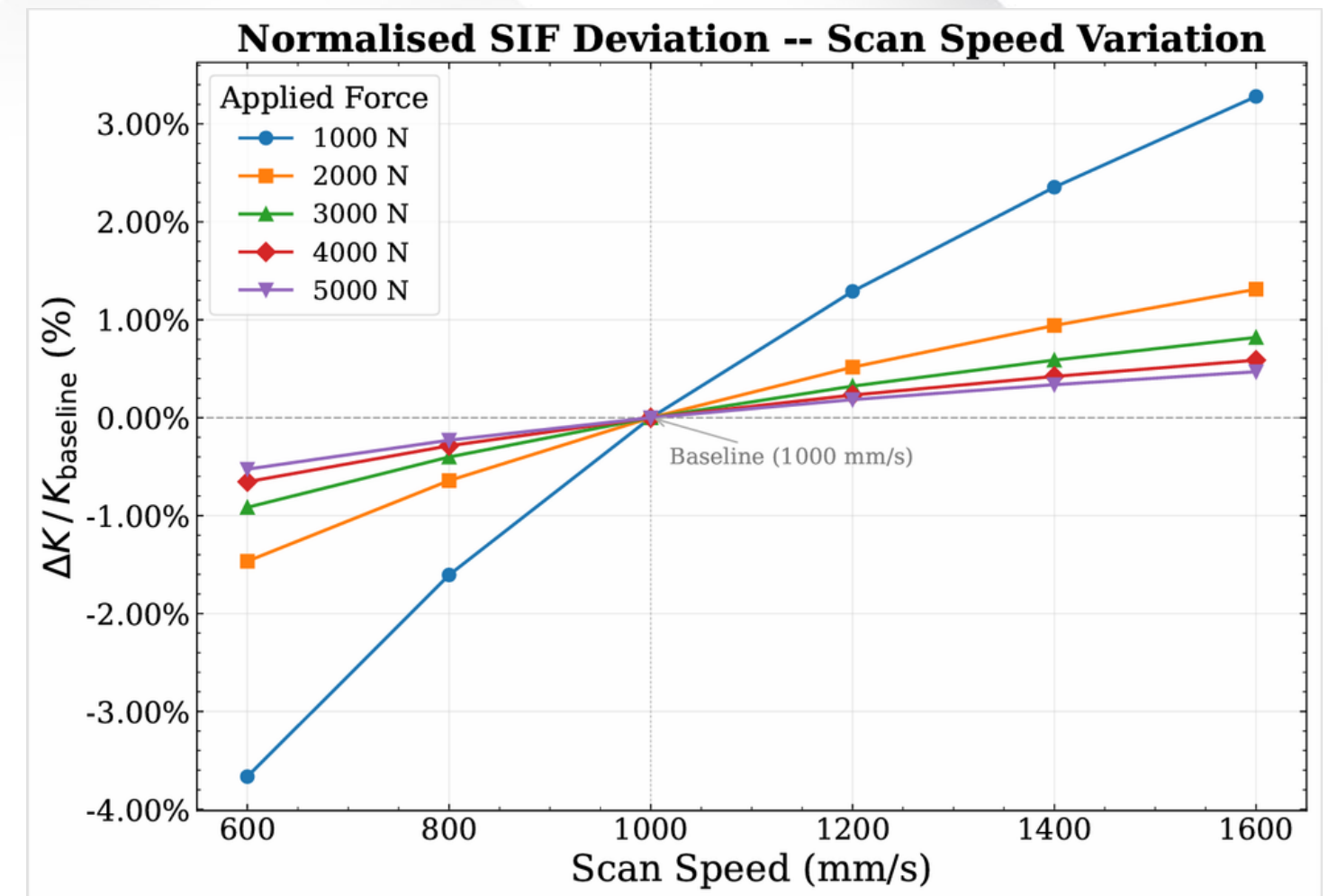
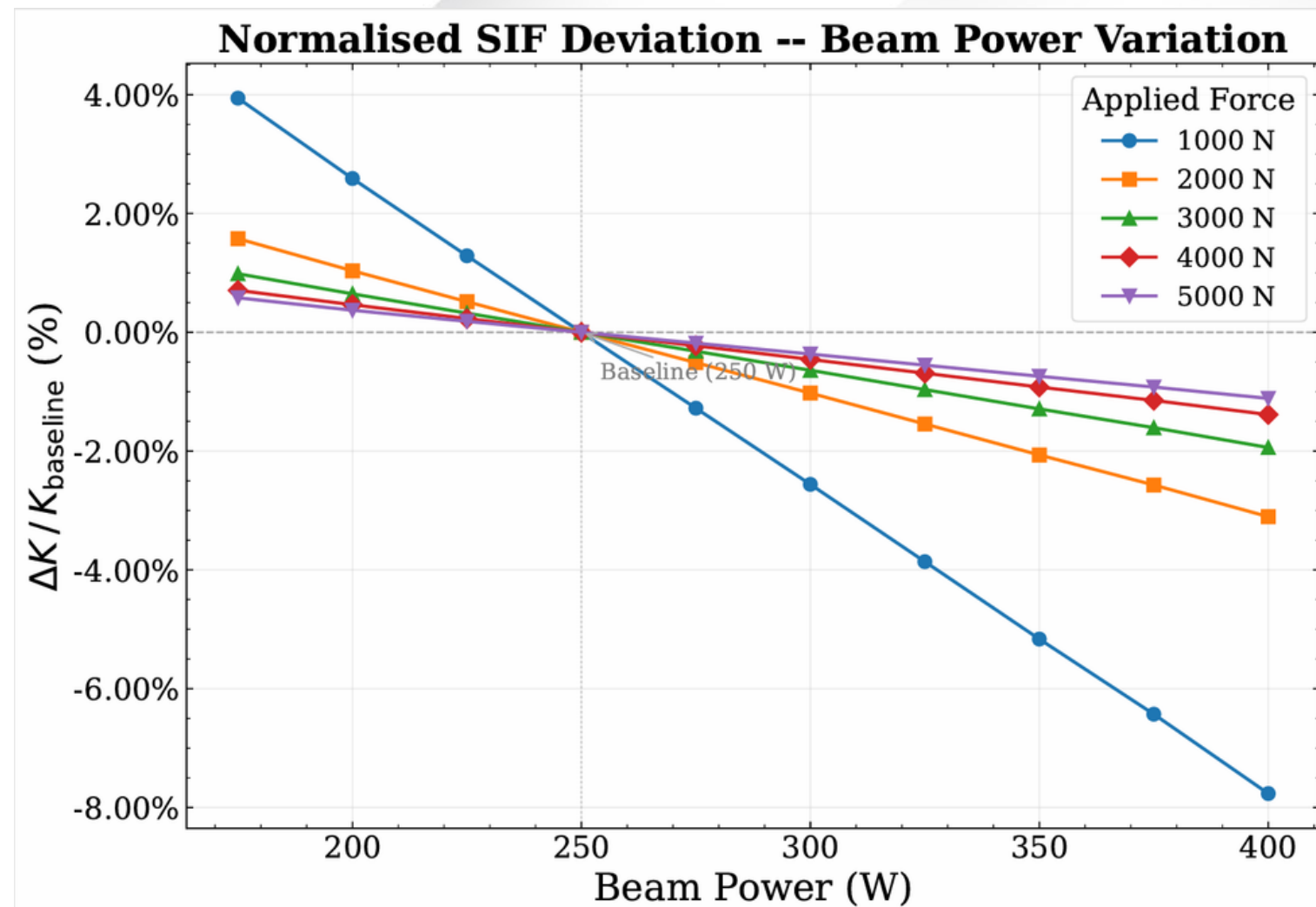
~5.33% variation.



SIF Stays same With layer Thickness : LT
Effects In Build Plane Direction

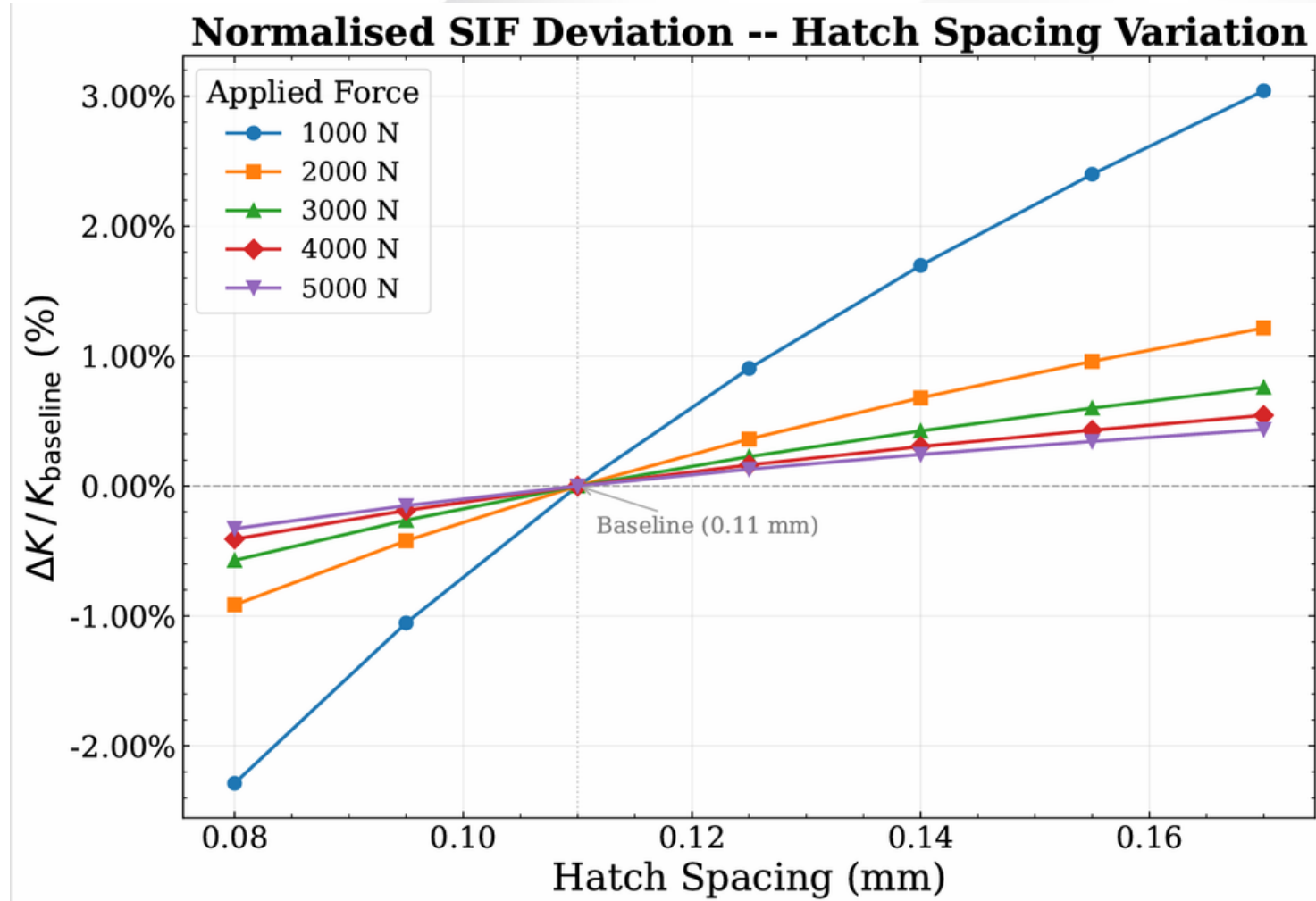
~0.07% variation.

SIF Results

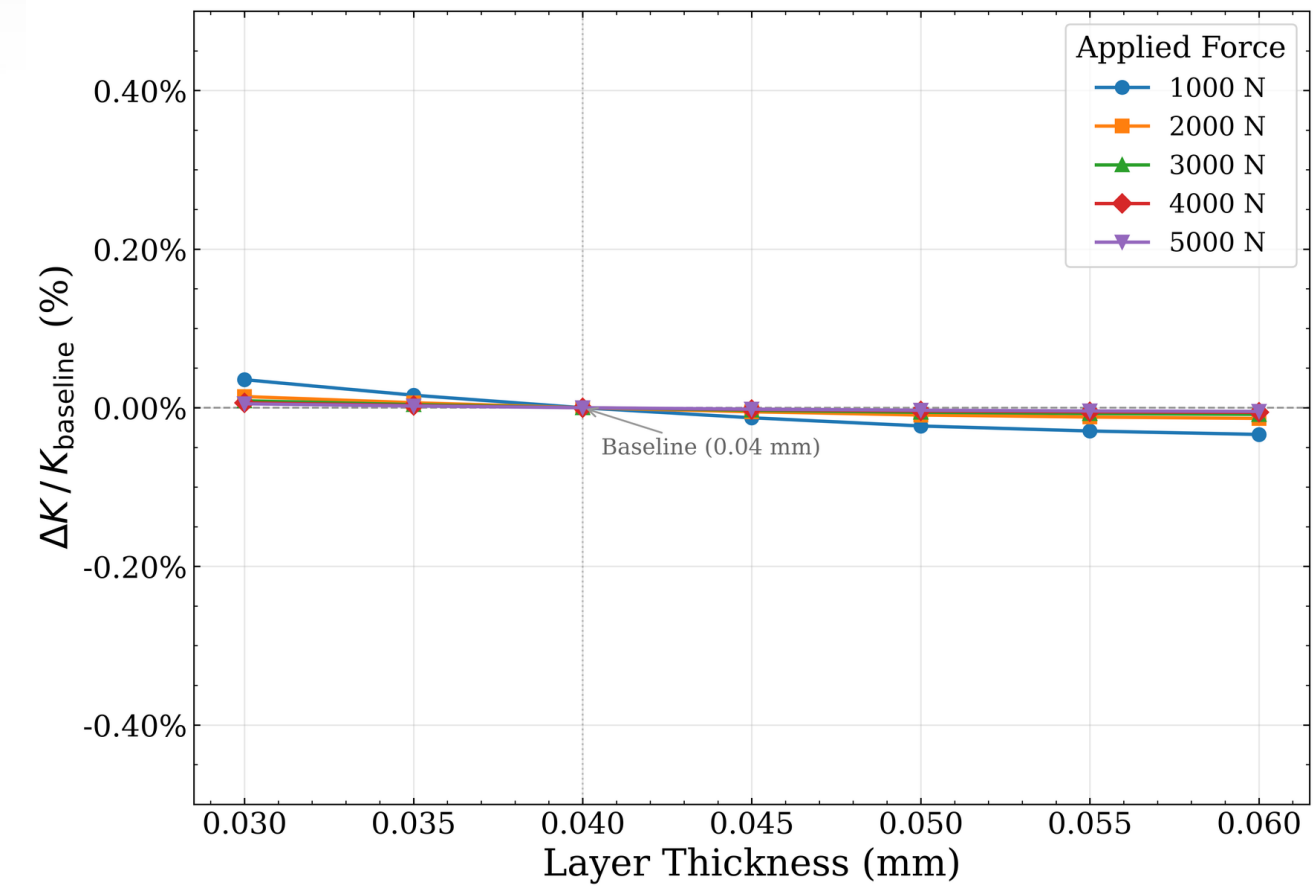


- Residual stress K contribution is a fixed offset
- Applied K scales linearly with force
- At high applied force, the externally driven component dominates and the relative influence of the residual stress diminishes

SIF Results

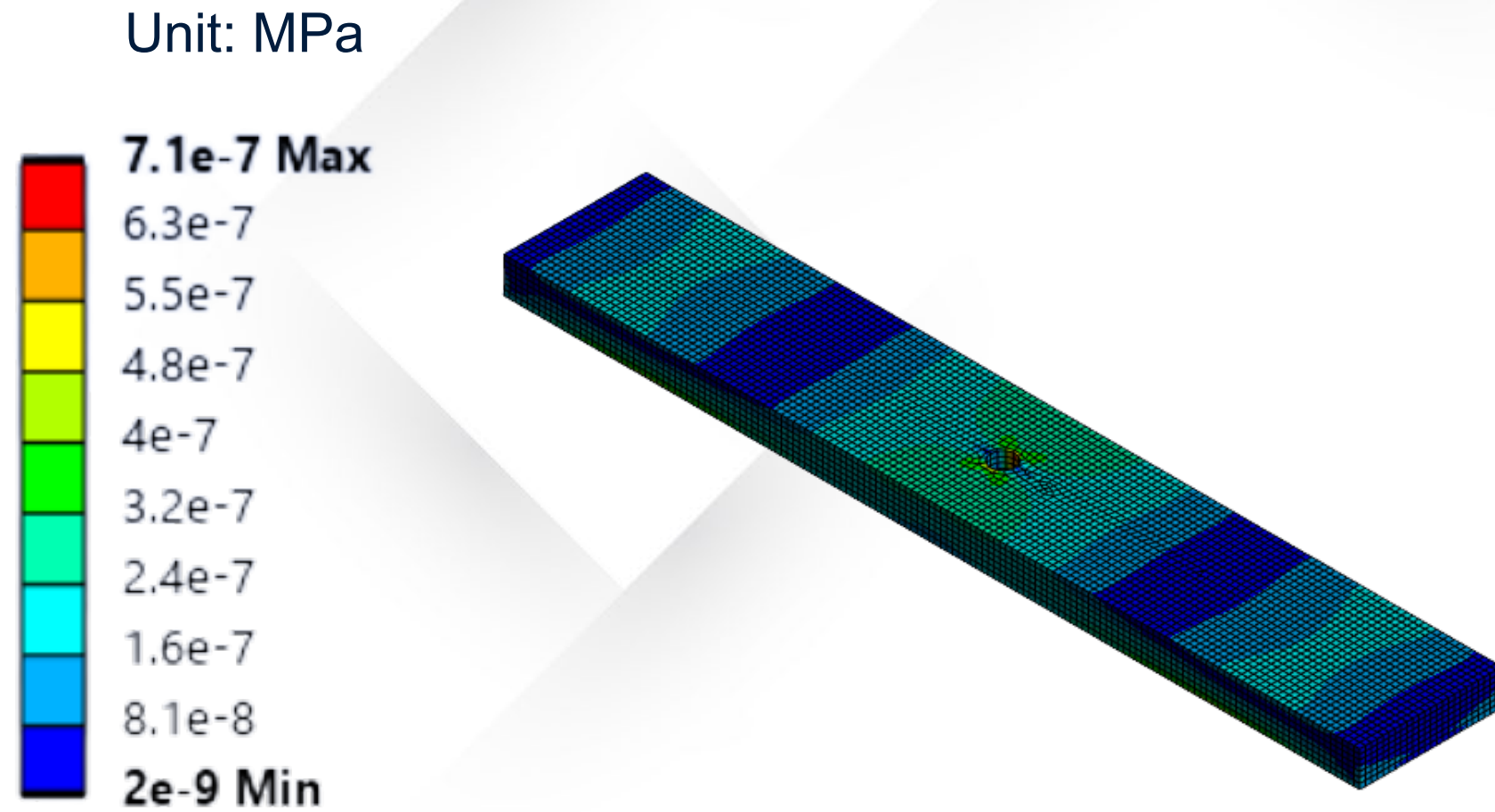


Normalised SIF Deviation -- Layer Thickness Variation

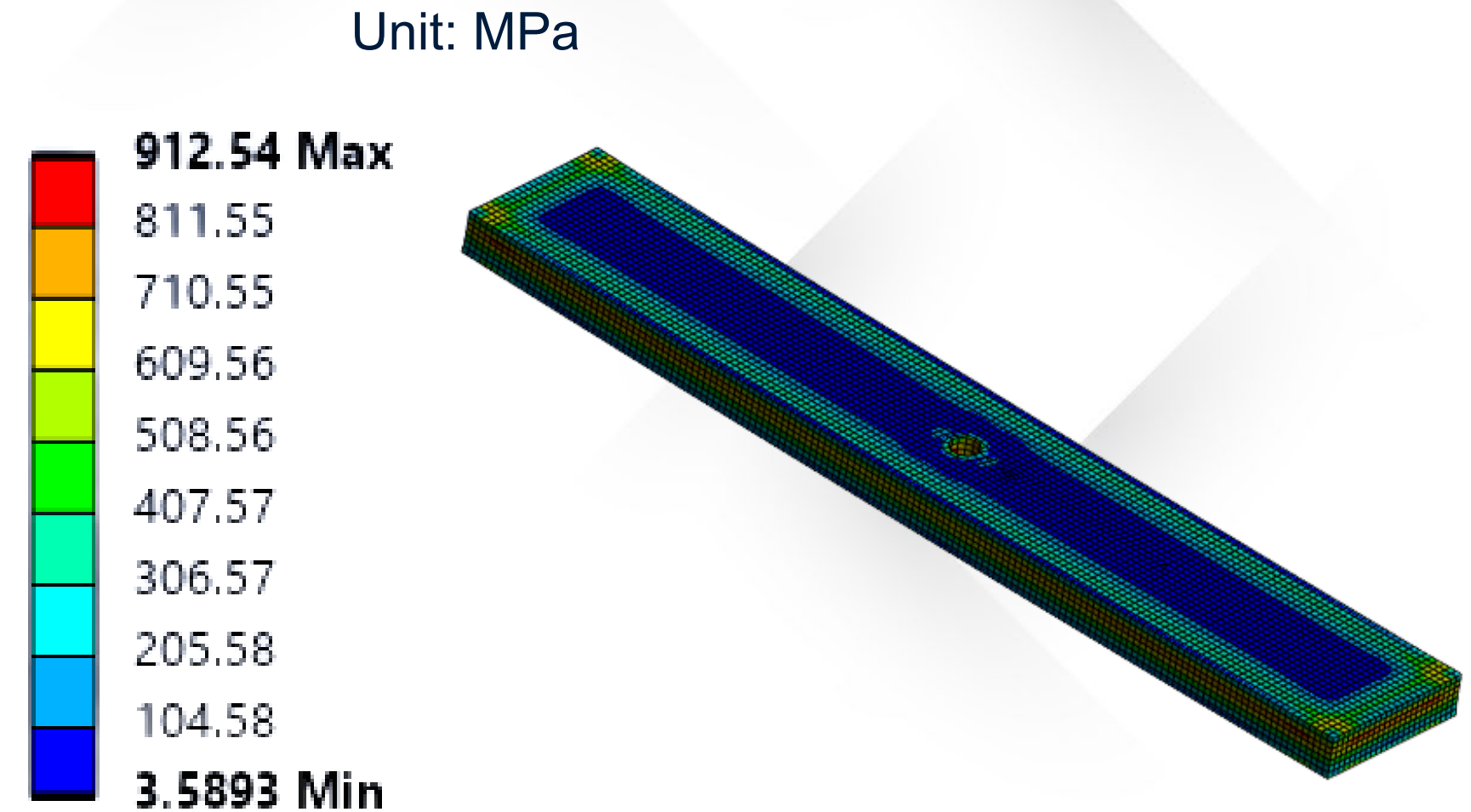


- Residual stress K contribution is a fixed offset
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Heat Treatment Effect



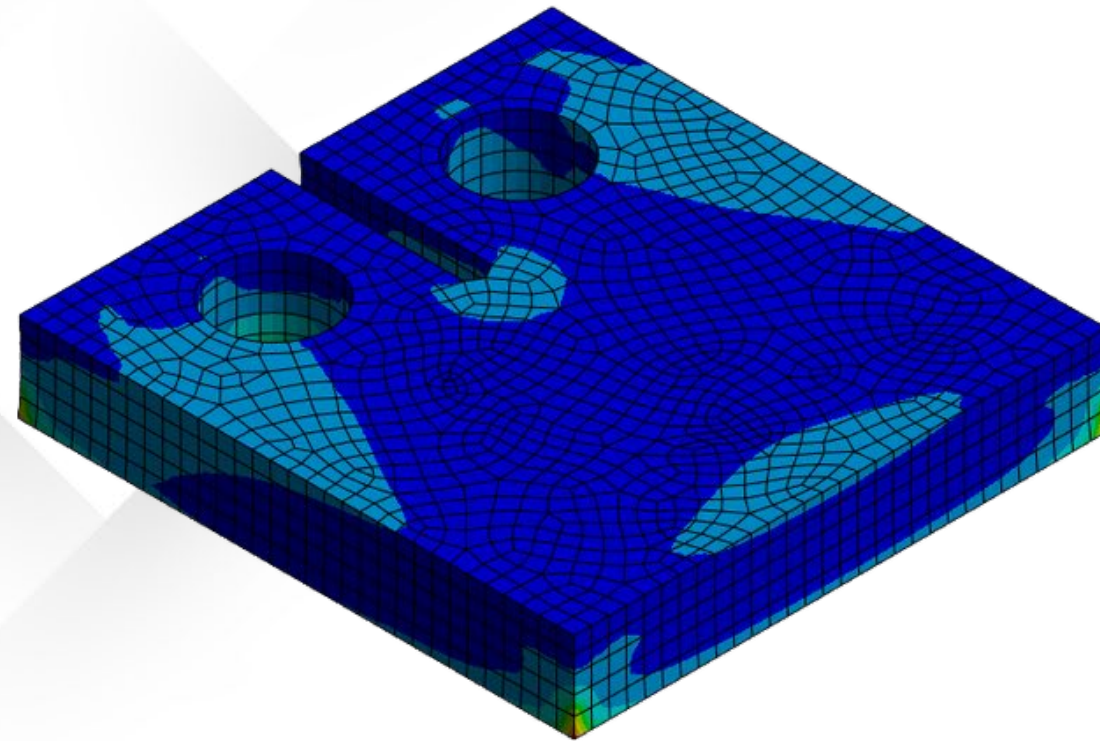
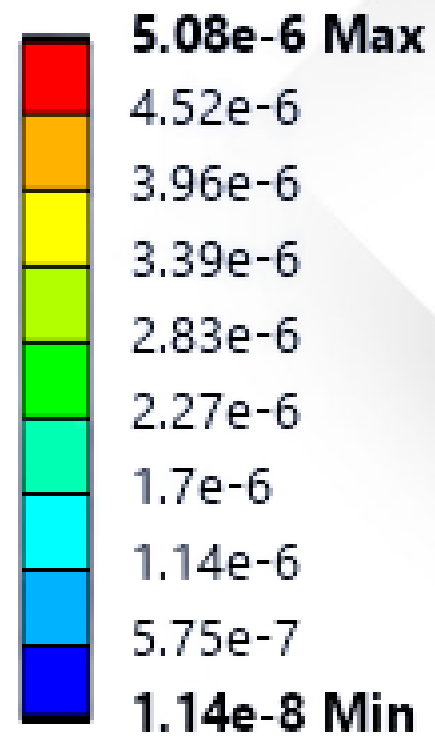
- Heat Treated Condition
- Elimination Of Residual Stress
- No Subsequent Effect On SCF & SIF



- As Built Condition
- Residual Stress Remains
- Subsequent Effect On SCF & SIF

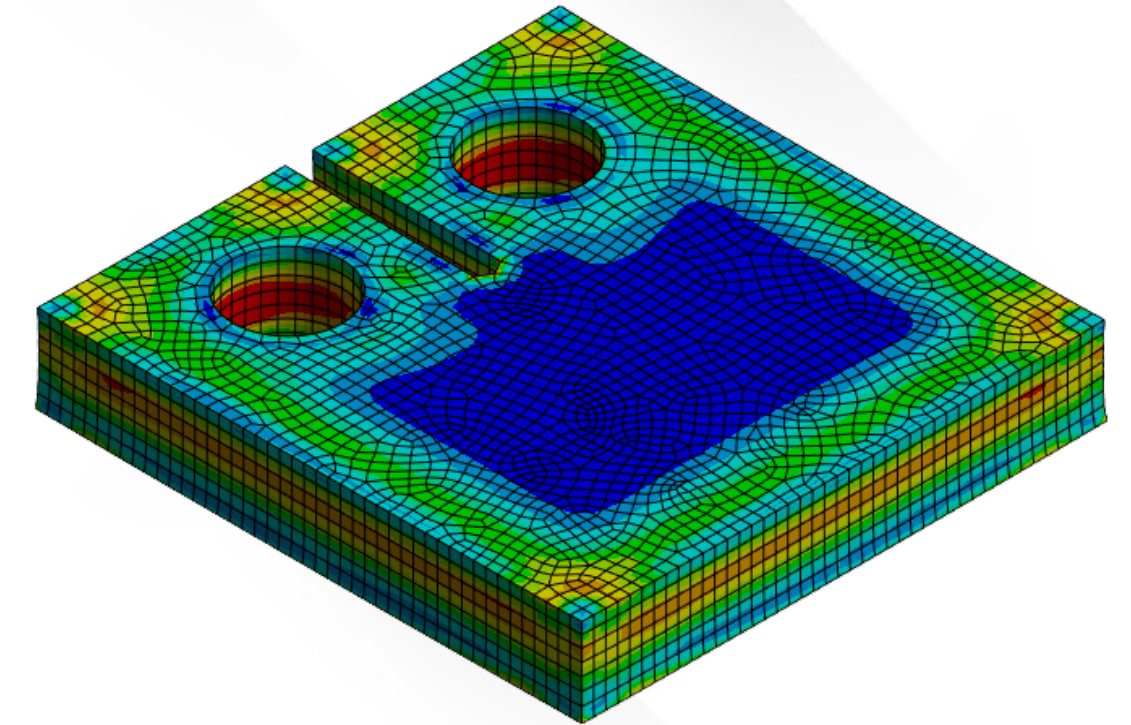
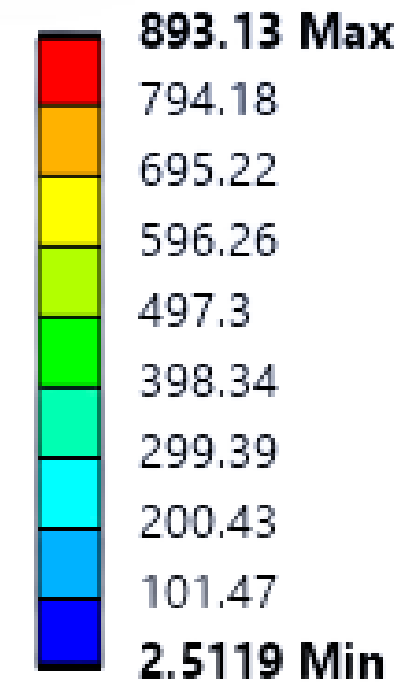
Heat Treatment Effect

Unit: MPa



- Heat Treated Condition
- Elimination Of Residual Stress
- No Subsequent Effect On SCF & SIF

Unit: MPa



- As Built Condition
- Residual Stress Remains
- Subsequent Effect On SCF & SIF

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Broader Impacts & Complex Engineering Problems

BROADER IMPACTS



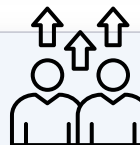
Health and Safety

Purely computational study; results guide safe, reliable physical fabrication downstream.



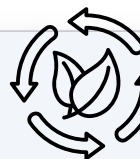
Standards & Qualification

Builds on ASTM E647, D5766; informs aerospace and nuclear part qualification.



Social & Economic

Lowers the experimental barrier in resource constrained settings; enables local high performance manufacturing.



Environment & Sustainability

Fewer physical trials mean less material and energy waste. Supports SDG 9, 12 and 13.

CEP Attribute Mapping

Confirmed when mandatory P1 holds, plus some or all of P2–P7.

P	Attribute	Status
P1	Depth of knowledge	✓
P2	Conflicting requirements	✓
P3	Depth of analysis	✓
P4	Familiarity of issues	Partial
P5	Applicable codes (ASTM)	✓
P6	Stakeholder involvement	✓

P1 + all six others met → CEP CONFIRMED

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Conclusions & Future Work

92–97%

of peak stress at the 2500 N service load is driven by residual stress for SCF

11.71%

K_I variation driven by beam power — the dominant process parameter

< 0.07%

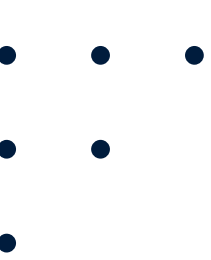
K_I variation from layer thickness — effectively no fracture-mechanical effect

KEY CONCLUSIONS

- Volumetric energy density ($VED = P / v \cdot h \cdot t$) is the single unifying driver behind every parameter trend.
- Sensitivity ranking: beam power (11.71%) > scan speed (6.95%) > hatch spacing (5.33%) $\gg$ layer thickness (< 0.07%). Layer thickness can be chosen on other manufacturing grounds
- The force- K_I relationship is strictly linear — parameters shift only the zero-load intercept (the residual-stress contribution), not the specimen compliance. Sensitivity is largest at low load; all results are as-built, and stress relief would remove the effect.

FUTURE WORK

- Experimental validation: printed specimens, residual-stress measurement, and fracture testing.
- Residual-stress redistribution during fatigue crack propagation
- Multi-parameter interaction via a full-factorial design of experiments



THANK

YOU

Q&A

